

APPENDIX F

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Application Number: 15826939

Document Date: 11/30/2017

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Application Number	
Filing Date	
First Named Inventor	Emre Alptekin
Title	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME
Art Unit	
Examiner Name	
Attorney Docket Number	FIS920150066US2

SIGNATURE of Applicant or Patent Practitioner

Signature	/Justine M. Wilbur/	Date (Optional)	2017-11-30
Name	Justine M. Wilbur	Registration Number	59,678
Title (if Applicant is a juristic entity)	Patent Practitioner for Applicant		
Applicant Name (if Applicant is a juristic entity)	GLOBALFOUNDRIES INC.		

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GLOBALFOUNDRIES INC.

- ☐ Inventor or Joint Inventor (title not required below)
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)
- ☒ Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)
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SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).

Signature		Date (Optional)	
Name	David Cain		
Title	Senior Counsel, IP		

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		
	Filing Date		
	First Named Inventor	Emre Alptekin	
	Art Unit		
	Examiner Name		
	Attorney Docket Number	FIS920150066US2	

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STATEMENT BY APPLICANT**
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Application Number	
Filing Date	
First Named Inventor	Emre Alptekin
Art Unit	
Examiner Name	
Attorney Docket Number	FIS920150066US2

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**INFORMATION DISCLOSURE
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Application Number	
Filing Date	
First Named Inventor	Emre Alptekin
Art Unit	
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Attorney Docket Number	FIS920150066US2

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Signature	/Justine M. Wilbur/	Date (YYYY-MM-DD)	2017-11-30
Name/Print	Justine M. Wilbur	Registration Number	59,678

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Application Number:				
Filing Date:				
Title of Invention:	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME			
First Named Inventor/Applicant Name:	Emre Alptekin			
Filer:	Justine M. Wilbur			
Attorney Docket Number:	FIS920150066US2			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
UTILITY APPLICATION FILING	1011	1	280	280
UTILITY SEARCH FEE	1111	1	600	600
UTILITY EXAMINATION FEE	1311	1	720	720
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1600

Electronic Acknowledgement Receipt

EFS ID:	31069961
Application Number:	15826939
International Application Number:	
Confirmation Number:	6013
Title of Invention:	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME
First Named Inventor/Applicant Name:	Emre Alptekin
Customer Number:	133964
Filer:	Justine M. Wilbur
Filer Authorized By:	
Attorney Docket Number:	FIS920150066US2
Receipt Date:	30-NOV-2017
Filing Date:	
Time Stamp:	11:37:58
Application Type:	Utility under 35 USC 111(a)

Payment information:

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Payment Type	CARD
Payment was successfully received in RAM	\$ 1600
RAM confirmation Number	113017INTEFSW11384600
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1		FIS920150066US2_FinalApplica tion_11-30-17.pdf	82343	yes	19
			88855e1272a373158520c6bc73e8c77bb5c e4865		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Abstract		19	19	
	Claims		17	18	
	Specification		1	16	
Warnings:					
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2	Drawings-only black and white line drawings	FIS920150066US2_FinalDrawin gs_11-30-17.pdf	110812	no	5
			b85ef96720d9772adb03c83feacf649d847 67d1		
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3	Application Data Sheet	FIS920150066US2_Application DataSheet_11-30-17.pdf	1823256	no	9
			c4a784a77382ce1e6bd98cb2a9111951242 352a0		
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4	Power of Attorney	FIS920150066US2_POA_11-30- 17.pdf	240668	no	2
			974fbc40baac65b710975a00d71213a5220 d7cb1		
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5	Information Disclosure Statement (IDS) Form (SB08)	FIS920150066US2_IDSPTOSB08 _11-30-17.pdf	1034987	no	4
			df651dc92bdd82fa657e43968b423dc44c8 b740		

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6	Fee Worksheet (SB06)	fee-info.pdf	35258	no	2
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ABSTRACT

An aspect of the invention includes a method for forming a contact in a dielectric layer over a semiconductor substrate. The method may comprise: forming a contact opening in a dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate; depositing a first liner layer to conformally coat the contact opening; causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate; depositing a refractory metal layer over the first intermix region; and depositing a metal in the contact opening thereby forming the contact.

CLAIMS

We claim:

1. A semiconductor structure including a contact to a semiconductor substrate, the contact comprising:

- a contact opening in a dielectric layer over the semiconductor substrate;
- a first liner layer substantially coating a sidewall the contact opening;
- a refractory metal layer substantially coating the first liner layer in the contact opening;
- a metal substantially filling the contact opening; and
- an intermix region at an upper surface of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate.

2. The contact of claim 1, wherein the intermix region includes a titanium-based silicide.

3. The contact of claim 1, wherein the first liner layer includes titanium (Ti) and the refractory metal layer includes titanium nitride (TiN).

4. The contact of claim 1, wherein the first liner layer includes nickel platinum (NiPt) and the refractory metal layer includes a bilayer having a first refractory metal material including titanium (Ti) and a second refractory metal material including titanium (TiN), the first refractory metal material substantially separating the first liner layer and the second refractory metal

material, the second refractory metal material substantially separating the first refractory metal material and the metal.

5. The contact of claim 1, wherein the metal includes tungsten (W).

6. The contact of claim 1, wherein the intermix region has a thickness of approximately 3 nanometers to approximately 5 nanometers.

CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME

BACKGROUND

Technical Field

[0001] The present invention relates to contacts over a semiconductor substrate and methods of forming the same. More specifically, the present invention relates to forming an intermix region at an interface of a bottom portion of contacts and an upper portion of the semiconductor substrate, respectively.

Related Art

[0002] State of the art field effect transistors (FETs) can be fabricated by depositing a gate conductor over a gate dielectric and a semiconductor substrate. Generally, the FET fabrication process implements lithography and etching processes to define the gate structures. After providing the gate structures, source/drain extensions may be formed into a portion of the semiconductor substrate and on both sides of each gate structure by ion implantation.

Sometimes this implant is performed using a spacer to create a specific distance between the gate structure and the implanted junction. Source and drain may be doped with a p-type or n-type dopant. The term "p-type" refers to the addition of impurities to an intrinsic semiconductor that creates deficiencies of valence electrons. "N-type" refers to the addition of impurities that contributes free electrons to an intrinsic semiconductor. After formation of the transistor, a set of contact openings may be created to expose the source and drain. The contact openings may be coated with a Ti-based silicide liner. Subsequently, contact openings may be filled with a metal

material, such as W.

[0003] In a p-type FET (PFET), performance with a Ti-based silicide liner is a challenge for 14/10 nm silicon-on-insulator (14/10 SOI) technologies. In order to improve workfunction of Ti-based silicide in the source and drain of a PFET, a bilayer of NiPt and Ti has been employed to coat contact openings over the source and drain. However, this bilayer requires deposition of a relatively thick metal layer. As semiconductor devices get smaller, requiring a thick bilayer in the contact causes contact opening fill issues, such as an incomplete contact fill.

SUMMARY

[0004] A first aspect of the invention includes a method for forming a contact in a dielectric layer to a semiconductor substrate. The method comprises: forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate; depositing a first liner layer to conformally coat the contact opening; causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate; depositing a refractory metal layer over the first intermix region; and depositing a metal in the contact opening thereby forming the contact.

[0005] A method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain. The method comprises: forming a contact opening in the dielectric layer to expose an upper portion of each of the source and the drain; depositing a first liner layer to conformally coat each contact opening and causing a portion of the first liner layer to diffuse into the upper portion of

each of the source and the drain to form a first intermix region at the upper portion of the each of the source and the drain; depositing a refractory metal layer over the first intermix region; and depositing a metal to substantially coat the refractory metal layer and substantially fill each of the openings thereby forming the set of contacts.

[0006] A third aspect of the invention includes a semiconductor structure including a contact to a semiconductor substrate. The contact comprises: a contact opening in a dielectric layer over the semiconductor substrate; a first liner layer substantially coating a sidewall of the contact opening; a refractory metal layer substantially coating the first liner layer in the contact opening; a metal substantially filling the contact opening; and an intermix region at an upper portion of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The embodiments of this invention will be described in detail, with reference to the following figures, wherein like designations denote like elements, and wherein:

[0008] FIGS. 1-8 show cross-sectional views of a semiconductor structure undergoing processing according to embodiments of the invention.

[0009] FIG. 9 shows a cross-sectional view of another embodiment of the semiconductor structure according to embodiments of the invention.

DETAILED DESCRIPTION

[0010] Aspects of the present invention utilize an intermix concept as a template for formation of a contact to a semiconductor substrate of a semiconductor structure. In some embodiments, the semiconductor structure may be a transistor, e.g., PFET or n-type FET (NFET) having a gate, a source, and a drain as is known in the art. In other embodiments, semiconductor structure may be a wire, a FinFet, or a hetero-bipolar junction transistor (HBT) or other silicon based microelectronic devices. One aspect of the present invention involves depositing a first liner layer in a contact opening such that an intermix region is formed at an interface of the bottom portion of the contact opening and an upper portion of the semiconductor substrate thereunder, e.g., source and drain. Unreacted (non-diffused) portions of the first liner layer may be removed prior to completion of fabricating the contact. Therefore, aspects of the present invention do not increase the deposited metal thickness in the contact openings. Thus, contact opening fill issues are avoided and overall contact resistance is improved.

[0011] Referring now to FIG. 1, a semiconductor structure 100 is shown including a semiconductor substrate 110, an isolation region 112, and a gate structure 114 within a dielectric layer 120 over substrate 112. Substrate 112 may be employed as a semiconductor layer of a semiconductor-on-insulator substrate (not shown), or a bulk substrate (not shown) as understood in the art. Substrate 112 may include any now known or later developed semiconductor material including but not limited to silicon, germanium, silicon germanium, silicon carbide, and those consisting essentially of one or more III-V compound semiconductors having a composition defined by the formula $Al_{X1}Ga_{X2}In_{X3}As_{Y1}P_{Y2}N_{Y3}Sb_{Y4}$, where $X1$, $X2$, $X3$, $Y1$, $Y2$, $Y3$, and $Y4$ represent relative proportions, each greater than or equal to zero and

$X1+X2+X3+Y1+Y2+Y3+Y4=1$ (1 being the total relative mole quantity). Other suitable substrates include II-VI compound semiconductors having a composition $Zn_{A1}Cd_{A2}Se_{B1}Te_{B2}$, where A1, A2, B1, and B2 are relative proportions each greater than or equal to zero and $A1+A2+B1+B2=1$ (1 being a total mole quantity). Furthermore, a portion or entire substrate may be strained. Substrate 112 may be doped, undoped or contain doped and undoped regions therein.

[0012] In some embodiments of the invention, substrate 110 may include at least one isolation region 112 therein. Isolation region 112 may be a trench isolation region or a field oxide isolation region as is known in the art. A trench isolation region can be formed utilizing a conventional trench isolation process well known to those skilled in the art. For example, lithography, etching and filling of the trench with a trench dielectric such as an oxide may be used in forming the trench isolation region. Optionally, a liner may be formed in the trench prior to trench fill. After the trench is filled, a densification step may be performed as well as a planarization process. A field oxide isolation region may be formed utilizing a so-called local oxidation of silicon process. The at least one isolation region 112 may provide isolation between neighboring gate structures, typically required when the neighboring gate structures have opposite conductivities, i.e., n-type transistors and p-type transistors. As such, the at least one isolation region 112 can separate, for example, a first device region in which an n-type transistor device can be formed and second device region in which a p-type transistor device can be formed.

[0013] Further, FIG. 1 illustrates that a gate 114 may be formed on substrate 112. Gate structure 114 may include a gate dielectric material 116 and a gate conductor 118. Gate structure 114 may be used to control output current of the semiconducting device through electrical or magnetic fields. Gate dielectric 116 may include, but is not limited to: HfO_2 , ZrO_2 , La_2O_3 , Al_2O_3 ,

TiO₂, SrTiO₃, LaAlO₃, Y₂O₃, HfO_xN_y, ZrO_xN_y, La₂O_xN_y, Al₂O_xN_y, TiO_xN_y, SrTiO_xN_y, LaAlO_xN_y, Y₂O_xN_y, SiON, SiN_x, a silicate thereof, and an alloy thereof. Each value of x is independently from 0.5 to 3 and each value of y is independently from 0 to 2. In some embodiments, a multilayered gate dielectric structure comprising different gate dielectric materials, e.g., silicon dioxide, and a dielectric metal oxide can be formed and used as gate dielectric material 116. Gate conductor 118 may include, but is not limited to: doped polysilicon, doped silicon germanium, an elemental metal (e.g., tungsten, titanium, tantalum, aluminum, nickel, ruthenium, palladium and platinum), an alloy of at least two elemental metals, an elemental metal nitride (e.g., tungsten nitride, aluminum nitride, and titanium nitride), or multilayered combinations thereof.

[0014] Gate structure 114 may be formed by depositing gate dielectric 116 on substrate 112. As used herein, “depositing” may include any now known or later developed techniques appropriate for the material to be deposited including but are not limited to, for example: chemical vapor deposition (CVD), low-pressure CVD (LPCVD), plasma-enhanced CVD (PECVD), semi-atmosphere CVD (SACVD) and high density plasma CVD (HDPCVD), rapid thermal CVD (RTCVD), ultra-high vacuum CVD (UHVCVD), limited reaction processing CVD (LRPCVD), metalorganic CVD (MOCVD), sputtering deposition, ion beam deposition, electron beam deposition, laser assisted deposition, thermal oxidation, thermal nitridation, spin-on methods, physical vapor deposition (PVD), atomic layer deposition (ALD), chemical oxidation, molecular beam epitaxy (MBE), plating, evaporation. Additionally, subsequent to the deposition of gate dielectric 116, gate conductor 118 can be deposited on gate dielectric 116. The precise thickness of gate dielectric 116 and gate conductor 118 may vary widely with the intended application.

[0015] Gate dielectric 116 and gate conductor 118 can be patterned into a functional gate structure 114. Gate structure 114 can be formed by lithography and etching. Lithography can include forming a photoresist (not shown) on the topmost surface of gate conductor 118 exposing the photoresist to a desired pattern of radiation, and then developing the exposed photoresist with a resist developer to provide a patterned photoresist atop the gate conductor 118. At least one etch is then employed which transfers the pattern from the patterned photoresist into the various materials. As used herein, "etching" may include any now known or later developed techniques appropriate for the material to be etched including but not limited to, for example: isotropic etching, anisotropic etching, plasma etching, sputter etching, ion beam etching, reactive-ion beam etching and reactive-ion etching (RIE).

[0016] Still referring to FIG. 1, a spacer 122 may be formed on each sidewall of gate 114. Spacers 122 may be formed by depositing a spacer material and etching the spacer material. Spacers 122 may include a spacer dielectric material as is known in the art, such as, but not limited to: a dielectric oxide, dielectric nitride, and/or dielectric oxynitride. Additionally, a source 124a and drain 124b may be formed in a portion of substrate 110 on either side of gate 114. In some embodiments, a source extension region and a drain extension region (not separately shown) may be formed prior to forming spacer 122 utilizing an extension ion implantation process. As such, a portion of each of the source extension region and the drain extension region would be located beneath the spacer 122. After forming the spacers 122, source 124a and drain 124b are formed utilizing a source/drain ion implantation process. An activation anneal may follow the implantation processes. Source 124a (including the corresponding source extension region) and drain 124b (including the corresponding drain extension region) may be doped with a p-type or n-type dopant. The term "p-type" refers to the addition of impurities to an

intrinsic semiconductor that creates deficiencies of valence electrons. In a silicon-containing substrate, examples of p-type dopants, i.e., impurities, include, but are not limited to, boron, aluminum, gallium and indium. "N-type" refers to the addition of impurities that contributes free electrons to an intrinsic semiconductor. In a silicon-containing substrate, examples of n-type dopants, i.e., impurities, include, but are not limited to, antimony, arsenic and phosphorous. The concentration of dopants within source 124a (and the corresponding source extension region) and drain 124b (and the corresponding drain extension region) can be within ranges known in the art.

[0017] Still referring to FIG. 1, a dielectric layer 120 may be formed such that dielectric layer 120 substantially surrounds gate 114. In one embodiment of the invention, formation of dielectric layer 120 may include deposition of a dielectric material 126 and a dielectric cap material 128. While shown as two separate layers, it is to be understood that dielectric layer 120 is not limited to only two layers. Rather, in other embodiments of the invention, dielectric layer 120 may include a single layer or more than two separate layers. Dielectric material 126 may include, but is not limited to: silicon nitride (Si_3N_4), silicon oxide (SiO_2), fluorinated SiO_2 (FSG), hydrogenated silicon oxycarbide (SiCOH), porous SiCOH , boro-phospho-silicate glass (BPSG), silsesquioxanes, carbon (C) doped oxides (i.e., organosilicates) that include atoms of silicon (Si), carbon (C), oxygen (O), and/or hydrogen (H), thermosetting polyarylene ethers, SiLK (a polyarylene ether available from Dow Chemical Corporation), a spin-on silicon-carbon containing polymer material available from JSR Corporation, other low dielectric constant (<3.9) material, or layers thereof. The upper surface of gate 114 is exposed immediately after depositing dielectric material 126. In some embodiments, a planarization process or an etch back process follows the deposition of the dielectric material 126. As is illustrated, dielectric material 126 may have an upper surface that is coplanar with an upper surface of gate 114. Dielectric

material 126 may have a bottom surface that contacts an upper surface of source 124a, an upper surface of the drain 124b, and an upper surface of insulation regions 112. The thickness of the dielectric material 126 that can be employed in the present application may vary depending on the type of dielectric material employed as well as the method that was employed in forming the same. In one embodiment, dielectric material 126 has a thickness from 80 nm to 500 nm. Other thicknesses that are greater or lesser than the range provided above can also be used for dielectric material 126.

[0018] Subsequently, dielectric cap material 128 may be deposited on dielectric material 126. Dielectric cap material 128 may include an insulator material that differs from dielectric material 126. Examples of insulator materials that can be used as dielectric cap material 128 include, for example, silicon dioxide, silicon nitride and silicon oxynitride. Dielectric cap material 128 may have a thickness from 1 nm to 20 nm. Other thicknesses can also be used for the thickness of the dielectric cap material 128.

[0019] As shown in FIG. 2, a plurality of contact openings 130a, 130b may be formed in a dielectric layer 120 to substrate 110. That is, contact openings 130a, 130b may be formed through dielectric cap material 128 and dielectric material 126 to expose source 124a and drain 124b. For example, contact opening 130a exposes source 124a, while contact opening 130b exposes drain 124b. Contact openings 130a, 130b can be formed by lithography and etching. Portions of dielectric material 126 and dielectric cap material 128 remain on the upper surfaces of gate 114. As is shown in the drawings of the present application, a portion of dielectric cap material 128 that is located on the upper surface gate structure 114 may extend beyond the edges of gate structure 114 and beyond an outermost edge of each spacer 122. Further, dielectric cap material 128 and dielectric material 126 that are present above isolation regions 112 extend

beyond the outermost edges of isolation regions 112. In some embodiments, plurality of openings 130a, 130b have a width (measured from one exposed sidewall of one portion of dielectric material 126 to another exposed sidewall of another portion of dielectric material 126) of from 10 nm-50 nm. Other widths that are greater or lesser than the aforementioned widths can also be used in the present application.

[0020] As shown in FIG. 3, a first liner layer 132 may be deposited to conformally coat each contact opening 130a, 130b and substantially cover dielectric cap material 128. First liner layer 132 may be deposited using ionized PVD or radio frequency PVD (RFPVD) or other directional energetic thin film deposition techniques such as Ion beam assisted deposition, performed at room temperature or above. First liner layer 132 may include, but is not limited to, at least one of: NiPt, Ni, Ti, TiSi, Co, Er, Pt, and Yb. First liner layer 132 may be deposited such that a first intermix region 134a, 134b is formed at an interface of the bottom portion of each contact opening 130a, 130b and the upper portion of each of source 124a and drain 124b.

[0021] First intermix regions 134a, 134b may originate from the high energy of the incoming sputtered atoms (from first liner layer 132) which are thus pushed into the Si lattice of source 124a and drain 124b. Therefore, first intermix regions 134a, 134b may include a mixture of the material used for first liner layer 132 and a material used for the respective semiconductor substrate 110 thereunder. In the embodiment shown in FIG. 3, first intermix regions 134a, 134b may have a composition including a material used for first liner layer 132 and materials used for the respective source and drain 124a, 124b thereunder. While the Si lattice is a dense way to pack Si atoms, it has a relatively open structure due to its bonding being covalent. The first few metal atoms, arriving with relatively high energy, can be forced into the interstitial sites of the Si lattice to create a thin region (intermix regions 134a, 134b) of mixed Si and metal on the order of

3-5 nm in thickness, for example. In some cases, first intermix region 134a, 134b may have a relatively high Si content (greater than 50% Si in atomic composition). In other embodiments, the compositions of first intermix regions 134a, 134b may vary depending on one or more factors including, but not limited to: the type of material being deposited, the composition of source 124a and drain 124b, the energy of the atoms of the material being deposited, and/or the conditions of the deposition technique (temperature, pressure, reaction, atmosphere, etc.). In some embodiments, it may be desirable to select a target composition of first intermix regions 134a, 134b in view of one or more criteria, e.g., etch selectivity. First intermix region 134a, 134b may be amorphous and a subsequent anneal may crystalize intermix region 134a, 134b into a lower resistance layer as will be described herein.

[0022] First intermix regions 134a, 134b allow for formation of a very thin contact layer that is normally very difficult to control with standard sputtering tools. First intermix regions 134a, 134b are much more conformal on Si structures that are not planar. Additionally, upon annealing of first intermix regions 134a, 134b (which will be described further herein), very little if any metal diffusion is needed to form the low resistance contact layer which is often a source of defects in small devices. Encroachment of silicide under spacers 122 and encroachment on defective silicon areas may be substantially reduced or eliminated by this intermixing concept, thus increasing the yield of semiconductor structures or devices.

[0023] Referring now to FIG. 4, another step of the method according to embodiments of the invention is shown. In this step, first liner layer 132 is removed from each contact opening 130a, 130b as well as from the upper surface of dielectric cap material 128. That is, the remaining portion of first liner layer 132 which does not diffuse into the respective source 124a and drain 124b is removed. First liner layer 132 may be removed by etching using one of the various

etching techniques previously described. In some embodiments, first liner layer 132 may be removed by an aqua regia etch process. Aqua regia etching is a wet etch process and involves providing a solution of hydrochloric and nitric acids to the surfaces to be etched. First liner layer 132 may be removed such that intermix regions 134a, 134b are exposed at the bottom portion of contact openings 130a, 130b.

[0024] As shown in FIG. 5, another aspect of the method according to embodiments of the invention is shown to include depositing a refractory metal layer 140 to conformally coat each contact opening 130a, 130b and substantially cover dielectric cap material 128 and first intermix regions 134a, 134b at the bottom portion of each contact opening 130a, 130b. Refractory metal layer 140 may include, but is not limited to: W, Ti, Ta or Mo employed directly as a “metal”; a refractory metal employed as a constituent of a “compound” thereof, e.g., TiN, TaN, etc.; or even as a constituent of an “alloy” with another metal, e.g., titanium-tungsten. In some embodiments, as shown in FIG. 5, refractory metal layer 140 may include a bilayer of a first refractory material 142 and a second refractory material 144 such that first refractory metal material 142 substantially separates first liner layer 132 from second refractory metal layer 144. First and second refractory metal materials 142, 144 may include any of the refractory metal materials previously mentioned. For example, in some embodiments, first refractory metal material 142 may include Ti and second refractory metal material 144 may include TiN. In this example, first refractory metal material 142 may be deposited using RFPVD and second refractory metal material 144 may be deposited using ALD. While shown as a bilayer, it is to be understood that refractory metal layer 140 may include any number of layers of refractory metal materials without departing from aspects of the invention.

[0025] Referring now to FIG. 6, a metal 150 may be deposited over dielectric cap material 128

and within each contact opening 130a, 130b to substantially coat refractory metal layer 140 therein and substantially fill contact openings 130a, 130b. Alternatively, a plating process that fills each contact opening 130a, 130b from the bottom upwards can be used. In an embodiment of the invention where refractory metal layer 140 includes a bilayer, second refractory metal material 144 may substantially separate first refractory metal material 142 from metal layer 150. Metal layer 150 may include, but is not limited to: W, Co, Cu, and Al. As shown in FIG. 7, semiconductor structure 100 may undergo a planarization process such as, for example, chemical mechanical polishing (CMP) to create contacts 170a, 170b within contact openings 130a, 130b. During the planarization process, portions of refractory metal layer 140 and metal layer 150 that extend beyond the upper surface of dielectric cap material 128 are removed such that refractory metal layer 140 and metal layer 150 within contact openings 130a, 130b are coplanar with the upper surface of dielectric cap material 128.

[0026] In one example, contacts 170a and 170b may include NiPt as first liner layer 132, a Ti/TiN bilayer as refractory metal layer 140, and W as metal 150. In this embodiment, first intermix regions 134a, 134b may include a NiPt silicide and second intermix regions 154a, 154b may include a NiPtTi silicide. However, it is to be understood that other combinations of materials may be used to create contacts 170a, 170b without departing from aspects of the invention. For example, in another embodiment, first liner layer 132 may include Ni, refractory metal layer 140 may include a Ti/TiN bilayer, and metal 150 may include W. In this embodiment, first intermix regions 134a, 134b may include a Ni silicide and second intermix regions 154a, 154b may include a NiTi silicide.

[0027] As shown in FIG. 8, a laser anneal process may be performed such that a second intermix region 154a, 154b is formed at the interface of the bottom portion of contacts 170a,

170b and upper portion of source 124a and drain 124b. Second intermix regions 154a, 154b may have a thickness of 3-5 nm. The laser anneal process may transform the composition of first intermix region 134a, 134b (FIGS. 3-7) into a second composition resulting in second intermix region 154a, 154b in place of first intermix region 134a, 134b. That is, the laser anneal process may crystalize intermix regions 134a, 134b into a lower resistance layer, i.e. a second intermix region 154a, 154b. Second intermix regions 154a, 154b may have compositions including a mixture of the materials used for source 124a and drain 124b, first liner layer 132, and refractory metal layer 140. In an embodiment where Ti is used as refractory metal layer 140, second intermix region 154a, 154b may be a silicide, such as but not limited to: TiSi, NiSi, or NiPtTiSi. That is, during the laser anneal process, refractory metal 140 may diffuse into first intermix regions 134a, 134b (FIGS. 3-7) to create second intermix regions 154a, 154b. The laser anneal process may be performed at temperature of approximately 700°C-1100°C for between approximately 0.1 milliseconds to approximately 1 millisecond. More particularly, the laser anneal process may be performed at approximately 800°C for approximately 0.25 milliseconds. In some embodiments, the anneal temperature and duration may be dependent on the materials being used. In one embodiment, the anneal can utilize a laser having an energy from 500 watts to 2000 watts. In another embodiment, the anneal can be performed utilizing a raster scan beam size of from 1 cm long to 100 microns wide. In one example, the anneal can be performed utilizing a 10.6 micron CO₂ laser. In order to promote the efficiency of the CO₂ laser coupling with the semiconductor substrate, a laser with a shorter wavelength (e.g., InGaAs laser at 880 nm) can be used in conjunction with the CO₂ laser.

[0028] Referring now to FIG. 9, another embodiment of the invention is shown. In this embodiment, first liner layer 132 is not removed prior to depositing refractory metal layer 140.

In this embodiment, first liner layer 132 is deposited as described with respect to FIG. 3, such that the deposition of first liner layer 132 creates first intermix region 134a, 134b. However, after deposition of first liner layer 132, refractory metal layer 140 is deposited as described with respect to FIG. 5. While refractory metal layer 140 is shown as a bilayer, refractory metal layer 140 may include any number of layers of refractory metal materials, as previously discussed. Further, this embodiment also includes etching refractory metal layer 140 from the upper surface of dielectric cap layer 128 such that a portion of refractory metal layer 140 remains substantially coating the sidewalls of contact openings 130a, 130b and first intermix regions 134a, 134b at the bottom portion of contact openings 130a, 130b as described with respect to FIG. 6. Additionally, a metal layer 150 may be deposited and polished as described with respect to FIGS. 6-7. Lastly, this embodiment may also undergo the laser anneal process to create second intermix regions 154a, 154b as described with respect to FIG. 8. Second intermix regions 154a, 154b may include a mixture of the materials used for source and drain regions 124a, 124b, first liner layer 132, and refractory metal layer 140. This embodiment allows for a higher quantity of first liner layer 132 in contacts 170a, 170b due to the remaining portions of first liner layer 132 on sidewalls of contact openings 130a, 130b. Additionally, this embodiment results in a simpler process as the step of removing liner layer 132 is not performed, and therefore selectivity may not be a problem.

[0029] Contacts 170a, 170b of this embodiment may include NiPt as first liner layer 132, a Ti/TiN bilayer for refractory metal layer 140, and W as metal 150. In this embodiment, first intermix regions 134a, 134b may include NiPt silicide and second intermix regions 154a, 154b may include a NiPtTi silicide. Alternatively, first liner layer 132 may include Ti, refractory metal 140 may include TiN, and metal 150 may include W. In this embodiment, first intermix region

134a, 134b may include TiSi and second intermix region 154a, 154b may include TiSi with a higher concentration of Ti.

[0030] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0031] The descriptions of the various embodiments of the present invention have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

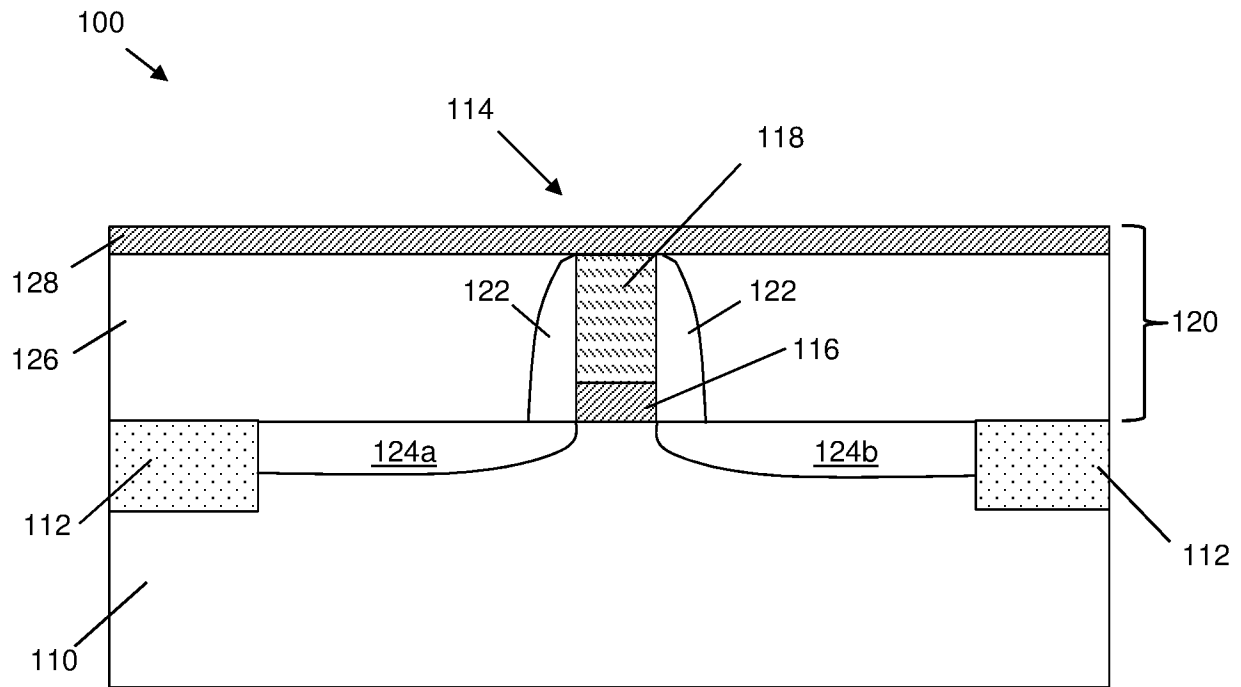


FIG. 1

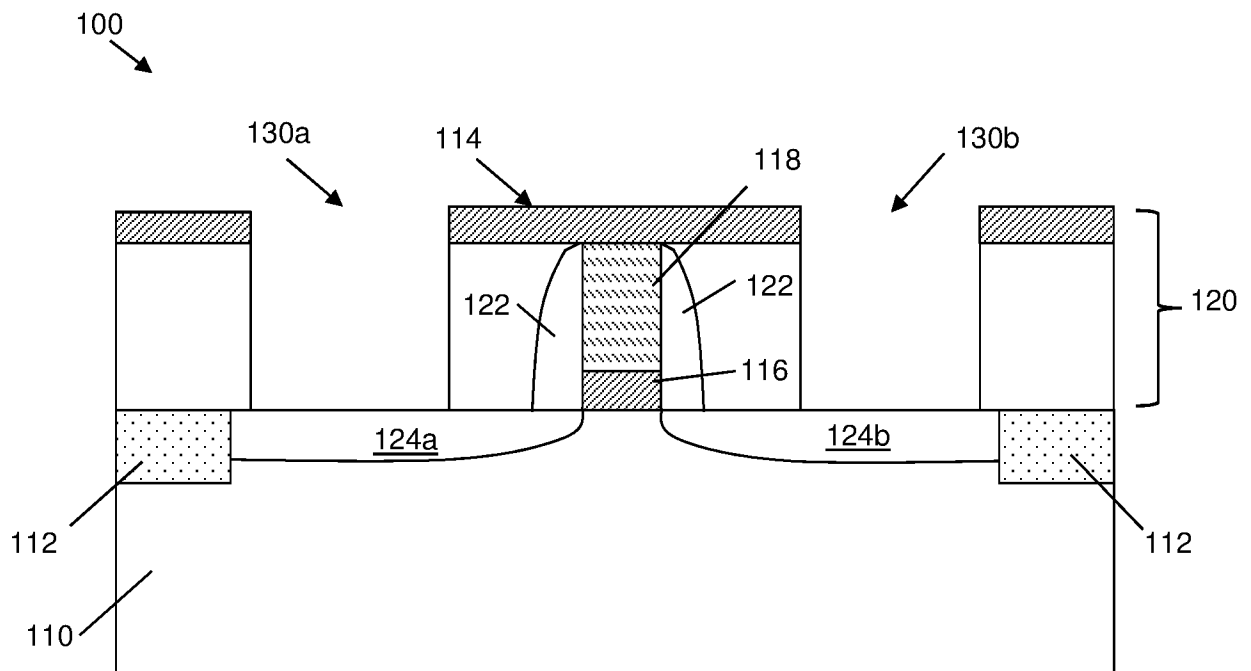


FIG. 2

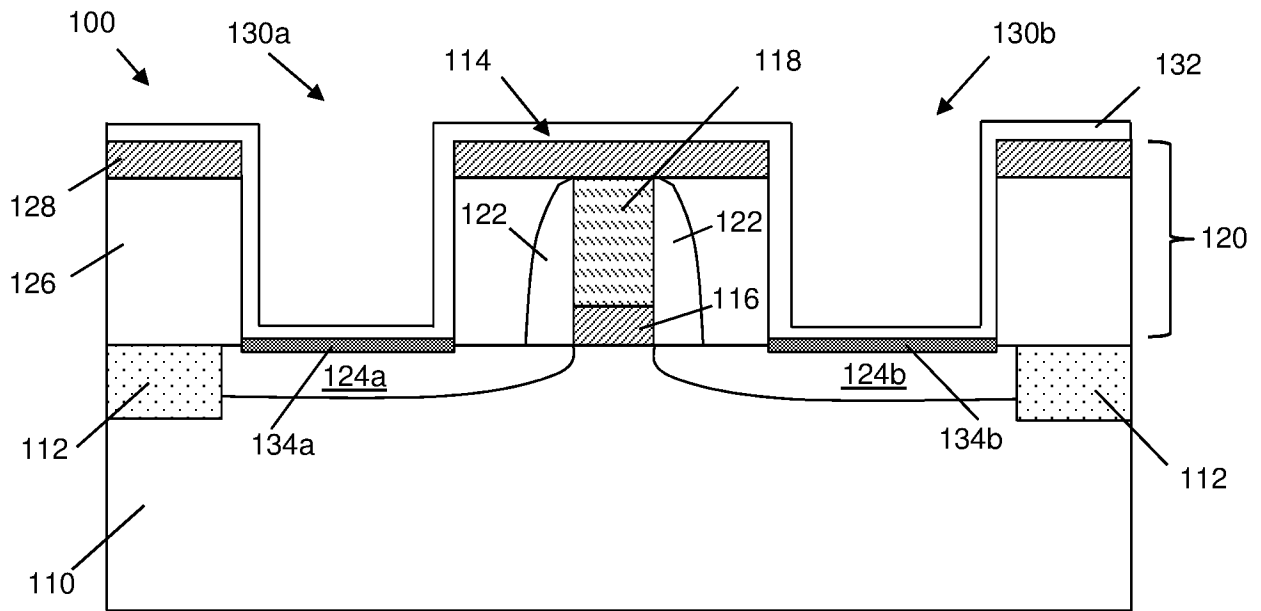


FIG. 3

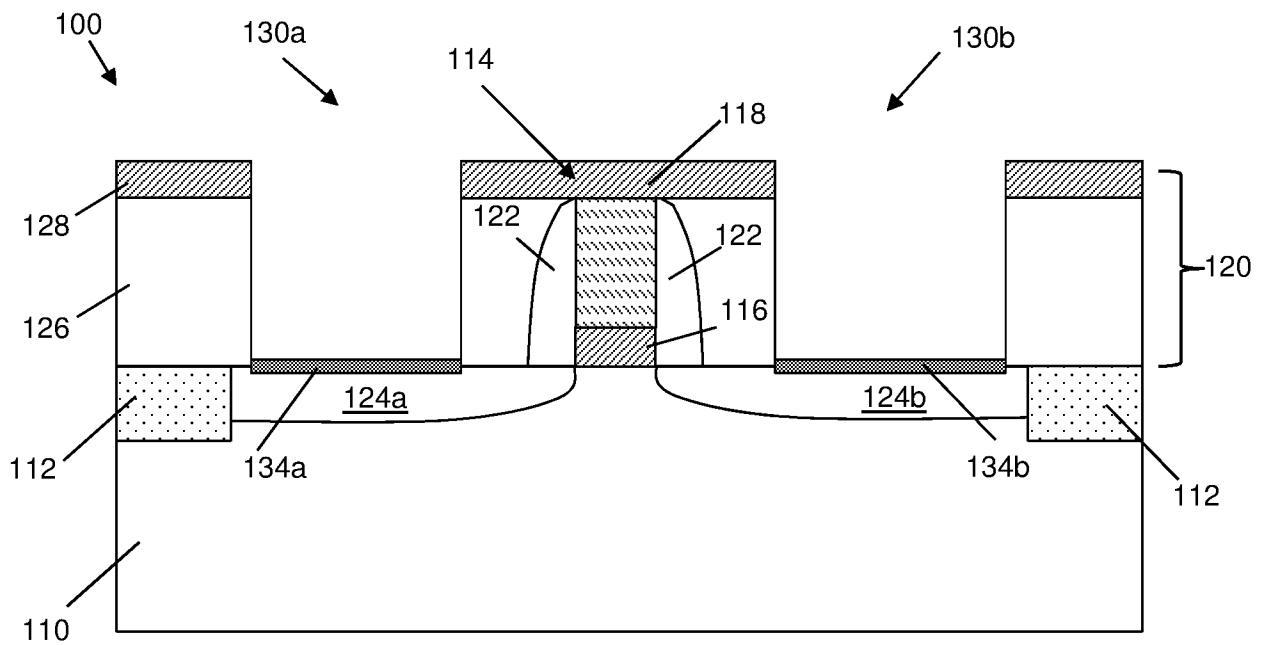


FIG. 4

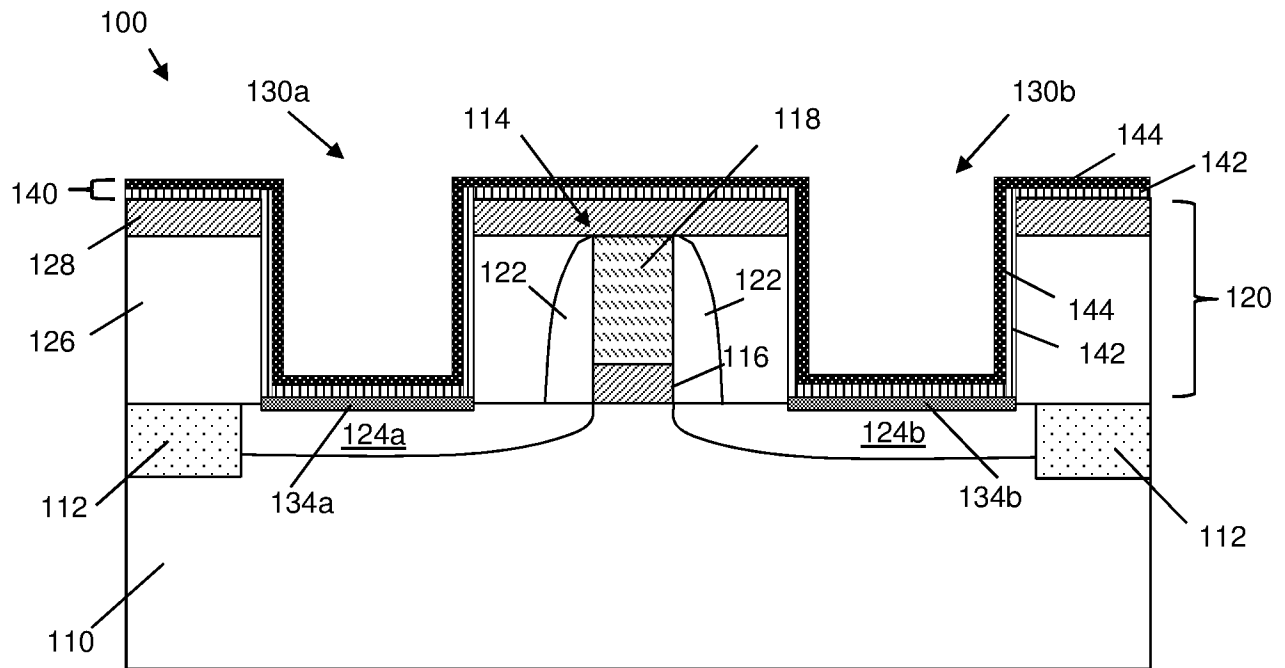


FIG. 5

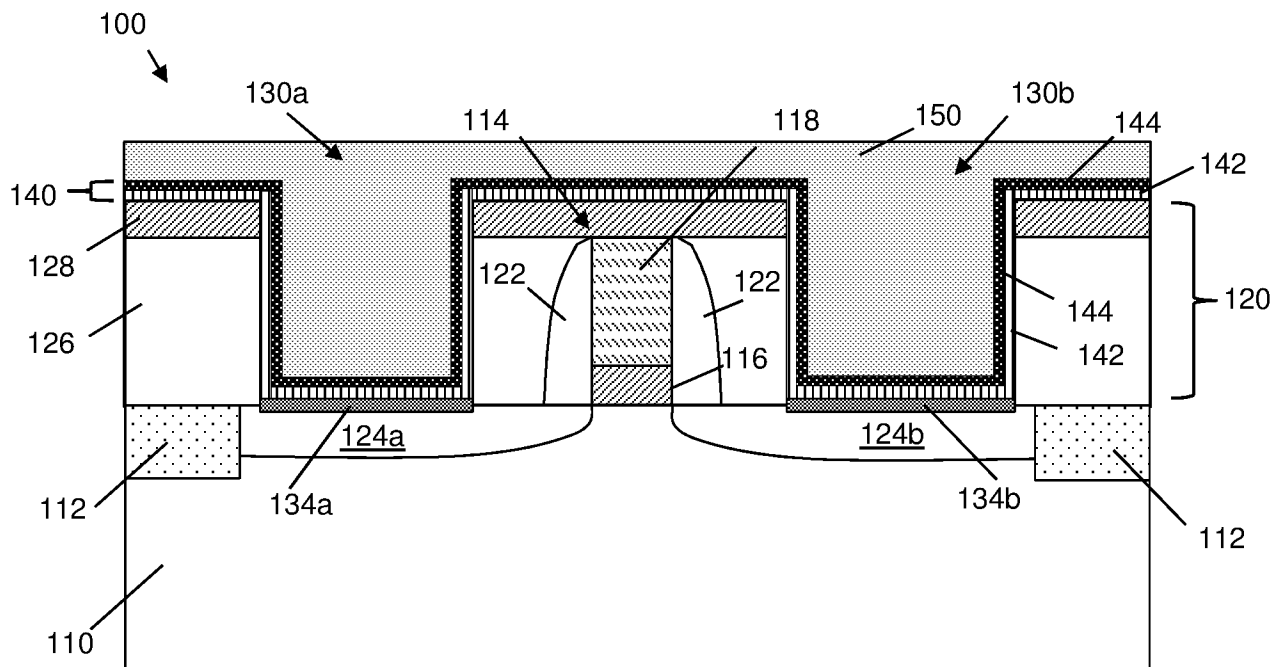


FIG. 6

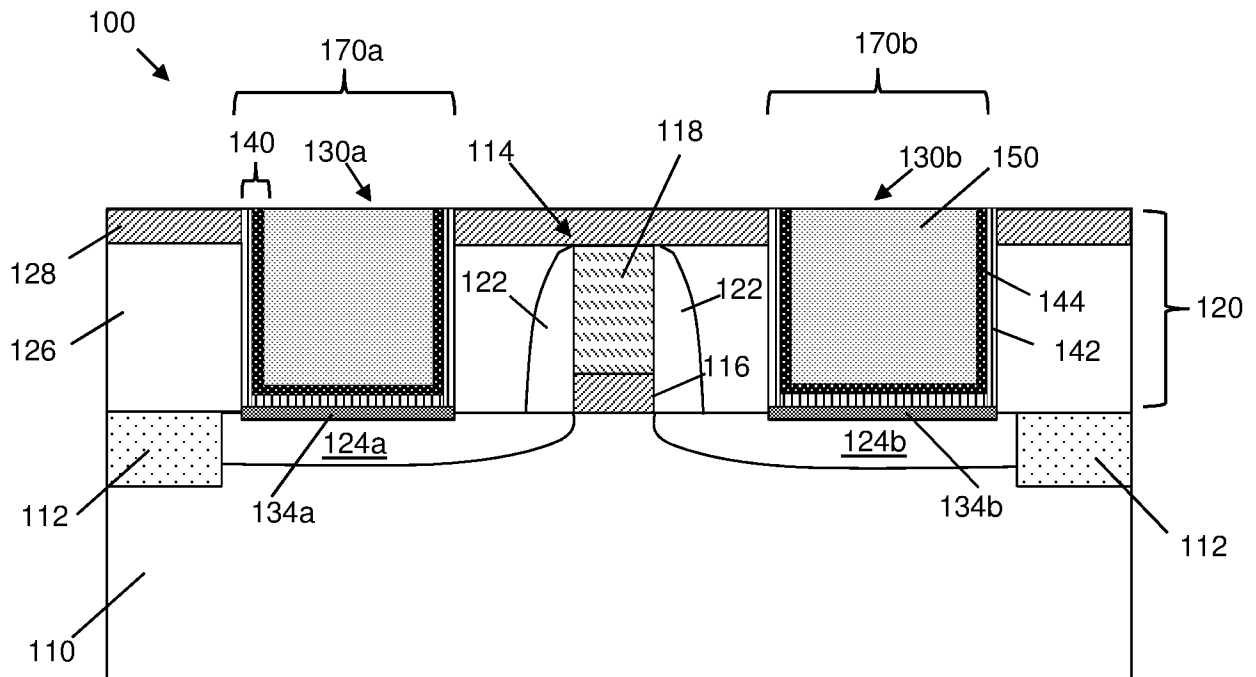


FIG. 7

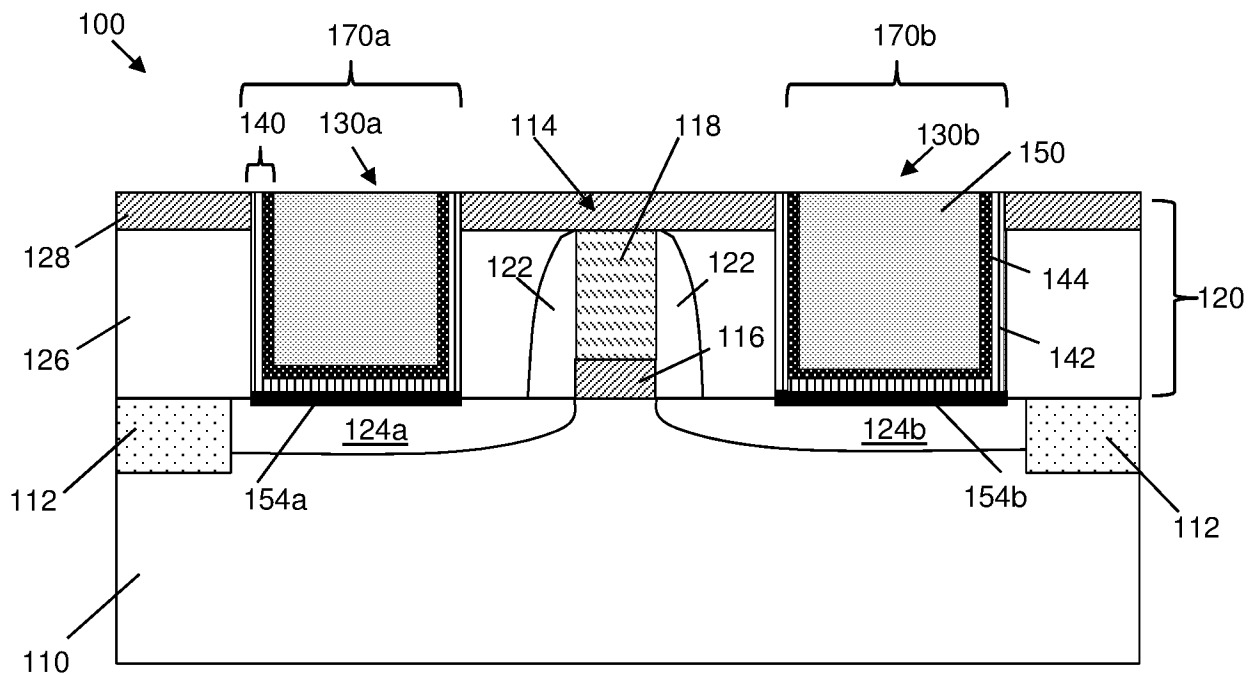


FIG. 8

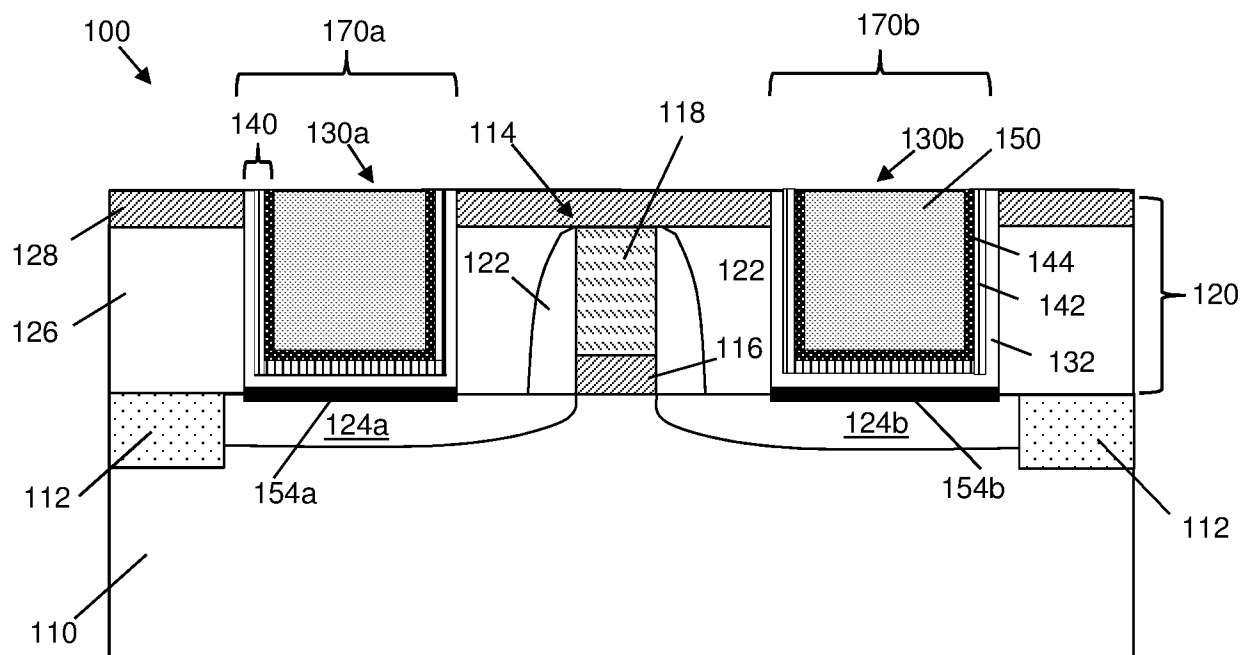


FIG. 9

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920150066US2
		Application Number	
Title of Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
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Prefix	Given Name	Middle Name	Family Name	Suffix	
	Emre		Alptekin		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Wappingers Falls	State/Province	NY	Country of Residence	US

Mailing Address of Inventor:

Address 1	2070 Route 52				
Address 2					
City	Hopewell Junction	State/Province	NY		
Postal Code	12533	Country	US		

Inventor	2				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Nicolas	L.	Breil		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	San Jose	State/Province	CA	Country of Residence	US

Mailing Address of Inventor:

Address 1	3401 Iron Point Drive				
Address 2					
City	San Jose	State/Province	CA		
Postal Code	95134	Country	US		

Inventor	3				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Christian		Lavoie		
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		Application Number	
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City	Pleasantville	State/Province	NY	Country of Residence	US
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Mailing Address of Inventor:

Address 1	1101 Kitchawan Road				
Address 2	Route 134				
City	Yorktown Heights	State/Province	NY		
Postal Code	10598	Country	US		
Inventor	4				Remove
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Prefix	Given Name	Middle Name	Family Name	Suffix	
	Ahmet	S.	Ozcan		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Chappaqua	State/Province	NY	Country of Residence	US

Mailing Address of Inventor:

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Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Wappingers Falls	State/Province	NY	Country of Residence	US

Mailing Address of Inventor:

Address 1	1101 Kitchawan Road				
Address 2					
City	Yorktown Heights	State/Province	NY		
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For further information see 37 CFR 1.33(a).

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920150066US2
		Application Number	
Title of Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME		

Customer Number	133964		
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Application Information:

Title of the Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME		
Attorney Docket Number	FIS920150066US2	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	5	Suggested Figure for Publication (if any)	

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☐ Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

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Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.			
Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	133964		

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920150066US2
		Application Number	
Title of Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME		

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This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
	Division of	14729298	2015-06-03
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.			Add

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			Remove
Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)
Additional Foreign Priority Data may be generated within this form by selecting the Add button.			Add

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.



NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76	Attorney Docket Number FIS920150066US2	
	Application Number	
Title of Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME	

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920150066US2
		Application Number	
Title of Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant	1	Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section. Clear		
☒ Assignee	Legal Representative under 35 U.S.C. 117	Joint Inventor
Person to whom the inventor is obligated to assign.		Person who shows sufficient proprietary interest
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:		
Name of the Deceased or Legally Incapacitated Inventor:		
If the Applicant is an Organization check here. ☒		
Organization Name	GLOBALFOUNDRIES INC.	
Mailing Address Information For Applicant:		
Address 1	P.O. Box 309	
Address 2	Ugland House	
City	Grand Cayman	State/Province
Country	KY	Postal Code
Phone Number		Fax Number
Email Address		
Additional Applicant Data may be generated within this form by selecting the Add button. Add		

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920150066US2
		Application Number	
Title of Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME		

Assignee 1				
Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.				
Remove				
If the Assignee or Non-Applicant Assignee is an Organization check here. ☐				
Prefix	Given Name	Middle Name	Family Name	Suffix
Mailing Address Information For Assignee including Non-Applicant Assignee:				
Address 1				
Address 2				
City		State/Province		
Country ⁱ		Postal Code		
Phone Number		Fax Number		
Email Address				
Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button. Add				

Signature:

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the **INITIAL** filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Justine M. Wilbur/		Date (YYYY-MM-DD)	2017-11-30
First Name	Justine	Last Name	Wilbur	Registration Number
				59,678
Additional Signature may be generated within this form by selecting the Add button. Add				

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920150066US2
		Application Number	
Title of Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

- 1 The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
- 2 A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
- 3 A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
- 4 A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
- 5 A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
- 6 A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
- 7 A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
- 8 A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
- 9 A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
15/826,939	11/30/2017	2812	1600	FIS920150066US2	6	1

CONFIRMATION NO. 6013

FILING RECEIPT



CC000000096152804

133964
HOFFMAN WARNICK LLC
540 Broadway
4th Floor
ALBANY, NY 12207

Date Mailed: 12/20/2017

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Emre Alptekin, Wappingers Falls, NY;
Nicolas L. Breil, San Jose, CA;
Christian Lavoie, Pleasantville, NY;
Ahmet S. Ozcan, Chappaqua, NY;
Kathryn T. Schonenberg, Wappingers Falls, NY;

Applicant(s)

GLOBALFOUNDRIES INC., Grand Cayman, CAYMAN ISLANDS;

Power of Attorney: The patent practitioners associated with Customer Number 133964

Domestic Priority data as claimed by applicant

This application is a DIV of 14/729,298 06/03/2015

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 12/18/2017

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 15/826,939**

Projected Publication Date: 03/29/2018

Non-Publication Request: No

Early Publication Request: No
Title

CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME

Preliminary Class

438

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
15/826,939

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	6 minus 20 =	*
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

OR

RATE(\$)	FEE(\$)
N/A	280
N/A	600
N/A	720
x 80 =	0.00
x 420 =	0.00
	0.00
	0.00
TOTAL	1600

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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United States Patent and Trademark Office
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P.O. Box 1450
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www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
15/826,939	11/30/2017	Emre Alptekin	FIS920150066US2

CONFIRMATION NO. 6013

FORMALITIES LETTER



133964
HOFFMAN WARNICK LLC
540 Broadway
4th Floor
ALBANY, NY 12207

Date Mailed: 12/20/2017

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing.

Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- Surcharge as set forth in 37 CFR 1.16(f) must be submitted.

The surcharge is due for any one of:

- late submission of the basic filing fee, search fee, or examination fee,
- late submission of inventor's oath or declaration,
- filing an application that does not contain at least one claim on filing, or
- submission of an application filed by reference to a previously filed application.

SUMMARY OF FEES DUE:

The fee(s) required within **TWO MONTHS** from the date of this Notice to avoid abandonment is/are itemized below. No entity status discount is in effect. If applicant is qualified for small entity status, a written assertion of small entity status must be submitted to establish small entity status. (See 37 CFR 1.27). If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- \$ 140 surcharge.
- \$(0) previous unapplied payment amount.
- \$ 140 TOTAL FEE BALANCE DUE.

Items Required To Avoid Processing Delays:

Applicant is notified that the above-identified application contains the deficiencies noted below. No period for reply is set forth in this notice for correction of these deficiencies. However, if a deficiency relates to the inventor's oath or declaration, the applicant must file an oath or declaration in compliance with 37 CFR 1.63, or a substitute statement in compliance with 37 CFR 1.64, executed by or with respect to each actual inventor no later than the expiration of the time period set in the "Notice of Allowability" to avoid abandonment. See 37 CFR 1.53(f).

- A properly executed inventor's oath or declaration has not been received for the following inventor(s):
Emre Alptekin
Nicolas L. Breil
Christian Lavoie
Ahmet S. Ozcan
Kathryn T. Schonenberg

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web, including a copy of this Notice and selecting the document description "Applicant response to Pre-Exam Formalities Notice".
<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at 1-866-217-9197 or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at **(571) 272-4000** or **(571) 272-4200** or **1-888-786-0101**.

/ngfissha/

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor: Emre Alptekin

Conf. No.: 6013

Serial No.: 15/826,939

Art Unit: 2812

Filed: November 30, 2017

Examiner: UNKNOWN

Docket No.: FIS920150066US2
(GFFT-0376-DIV)

Title: **CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME**

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

**RESPONSE TO NOTICE TO FILE MISSING PARTS
OF NONPROVISIONAL APPLICATION**

This paper is being filed in response to the Notice to File Missing Parts of NonProvisional Application dated December 20, 2017. Please consider the following remarks regarding the above-referenced patent application.

REMARKS

Inventor's Oath or Declaration

The Notice to File Missing Parts dated December 20, 2017 indicates that an inventor's oath or declaration has not been received for the five named inventors. In response, Applicant notes that the current application is a Divisional Application of U.S. App. No. 14/729,298 (filed June 3, 2015). Accordingly, and pursuant to 35 U.S.C. 115(g)(1)(A) quoted below, Applicant is not required to submit another copy of an inventor's oath or declaration if such was filed in connection with the earlier-filed application (in this case, parent application 14/729,298).

35 U.S.C. 115

(g) EARLIER-FILED APPLICATION CONTAINING REQUIRED STATEMENTS OR SUBSTITUTE STATEMENT.

(1) EXCEPTION. The requirements under this section shall not apply to an individual with respect to an application for patent in which the individual is named as the inventor or a joint inventor and that claims the benefit under section 120, 121, 365(c), or 386(c) of the filing of an earlier-filed application, if

(A) an oath or declaration meeting the requirements of subsection (a) was executed by the individual and was filed in connection with the earlier-filed application;

In light of 35 U.S.C. 115(g)(1)(A), the Office is directed to the parent application 14/729,298 for a copy of the executed oath/declaration for the five named inventors referenced in the Notice to File Missing Parts.

Surcharge

The Notice to File Missing Parts dated December 20, 2017 indicates that a surcharge is due for late submission of an inventor's oath or declaration. However, the surcharge as set forth

in 37 C.F.R. 1.16(f) for late submission of an inventor's oath or declaration is not believed to be due for the same reasons noted above with respect to the copy of the inventors' oaths or declarations.

No fees are believed due at this time. However, if any fees are required, the Office is hereby authorized to charge deposit account 500999.

Respectfully submitted,

/ Justine M. Wilbur /
Justine M. Wilbur
Reg. No. 59,678

Date: February 20, 2018

Hoffman Warnick LLC
540 Broadway, 4th Floor
Albany, New York 12207
Phone: (518) 449-0044
Fax: (518) 449-0047

Electronic Acknowledgement Receipt

EFS ID:	31827171
Application Number:	15826939
International Application Number:	
Confirmation Number:	6013
Title of Invention:	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME
First Named Inventor/Applicant Name:	Emre Alptekin
Customer Number:	133964
Filer:	Justine M. Wilbur
Filer Authorized By:	
Attorney Docket Number:	FIS920150066US2
Receipt Date:	20-FEB-2018
Filing Date:	30-NOV-2017
Time Stamp:	12:00:50
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Applicant Response to Pre-Exam Formalities Notice	FIS920150066US2_ResponseToNoticeToFileMissingParts_02-20-18.pdf	25703 10faed5a7fa793cb8e21d9d89576697ea9e7e31d	no	3

Warnings:

Information:	
Total Files Size (in bytes):	25703
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.	

Document code: WFEE

United States Patent and Trademark Office
Sales Receipt for Accounting Date: 02/23/2018

VVAN11	SALE	#00000007	Mailroom Dt: 02/20/2018	500999	15826939
		01 FC : 1051		160.00 DA	



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
15/826,939	11/30/2017	Emre Alptekin	FIS920150066US2

CONFIRMATION NO. 6013

INFORMAL NOTICE



133964
HOFFMAN WARNICK LLC
540 Broadway
4th Floor
ALBANY, NY 12207

Date Mailed: 02/26/2018

INFORMATIONAL NOTICE TO APPLICANT

Applicant is notified that the above-identified application contains the deficiencies noted below. No period for reply is set forth in this notice for correction of these deficiencies. However, if a deficiency relates to the inventor's oath or declaration, the applicant must file an oath or declaration in compliance with 37 CFR 1.63, or a substitute statement in compliance with 37 CFR 1.64, executed by or with respect to each actual inventor no later than the expiration of the time period set in the "Notice of Allowability" to avoid abandonment. See 37 CFR 1.53(f).

The item(s) indicated below are also required and should be submitted with any reply to this notice to avoid further processing delays.

- A properly executed inventor's oath or declaration has not been received for the following inventor(s):
Emre Alptekin
Nicolas L. Breil
Christian Lavoie
Ahmet S. Ozcan
Kathryn T. Schonenberg

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/sareebuddin/



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
15/826,939	11/30/2017	2897	1760	FIS920150066US2	6	1

133964
HOFFMAN WARNICK LLC
540 Broadway
4th Floor
ALBANY, NY 12207

CONFIRMATION NO. 6013
UPDATED FILING RECEIPT



Date Mailed: 02/26/2018

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Emre Alptekin, Wappingers Falls, NY;
Nicolas L. Breil, San Jose, CA;
Christian Lavoie, Pleasantville, NY;
Ahmet S. Ozcan, Chappaqua, NY;
Kathryn T. Schonenberg, Wappingers Falls, NY;

Applicant(s)

GLOBALFOUNDRIES INC., Grand Cayman, CAYMAN ISLANDS;

Power of Attorney: The patent practitioners associated with Customer Number 133964

Domestic Priority data as claimed by applicant

This application is a DIV of 14/729,298 06/03/2015 PAT 9865546

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 12/18/2017

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 15/826,939**

Projected Publication Date: 03/29/2018

Non-Publication Request: No

Early Publication Request: No
Title

CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME

Preliminary Class

257

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
15/826,939

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	6 minus 20 =	*
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

OR

RATE(\$)	FEE(\$)
N/A	300
N/A	660
N/A	760
x 100 =	0.00
x 460 =	0.00
	0.00
	0.00
TOTAL	1720

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
15/826,939	11/30/2017	Emre Alptekin	FIS920150066US2

CONFIRMATION NO. 6013

PUBLICATION NOTICE



OC000000098411641

133964
HOFFMAN WARNICK LLC
540 Broadway
4th Floor
ALBANY, NY 12207

Title:CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME

Publication No.US-2018-0090447-A1

Publication Date:03/29/2018

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be accessed through the USPTO's publically available Searchable Databases via the Internet at www.uspto.gov. The direct link to access the publication is currently <http://www.uspto.gov/patft/>.

The publication process established by the Office does not provide for mailing a copy of the publication to applicant. A copy of the publication may be obtained from the Office upon payment of the appropriate fee set forth in 37 CFR 1.19(a)(1). Orders for copies of patent application publications are handled by the USPTO's Public Records Division. The Public Records Division can be reached by telephone at (571) 272-3150 or (800) 972-6382, by facsimile at (571) 273-3250, by mail addressed to the United States Patent and Trademark Office, Public Records Division, Alexandria, VA 22313-1450 or via the Internet.

In addition, information on the status of the application, including the mailing date of Office actions and the dates of receipt of correspondence filed in the Office, may also be accessed via the Internet through the Patent Electronic Business Center at www.uspto.gov using the public side of the Patent Application Information and Retrieval (PAIR) system. The direct link to access this status information is currently <https://portal.uspto.gov/pair/PublicPair>. Prior to publication, such status information is confidential and may only be obtained by applicant using the private side of PAIR.

Further assistance in electronically accessing the publication, or about PAIR, is available by calling the Patent Electronic Business Center at 1-866-217-9197.

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

**DECLARATION (37 C.F.R. 1.63) FOR UTILITY PATENT APPLICATION USING AN
APPLICATION DATA SHEET (37 C.F.R. 1.76) AND ASSIGNMENT**

Title of Invention: **CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING
SAME**

As a below named inventor, I hereby declare that:

This declaration is directed to the attached application, or (if following box is checked):

☒ United States application or PCT international application number 14/729,298 filed on
June 3, 2015.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I have reviewed and understand the contents of the application, including the claims.

I am aware of the duty to disclose to the United States Patent and Trademark Office all information known to me to be material to patentability as defined in 37 CFR Section 1.56.

Whereas, the undersigned inventor(s) has/have made certain inventions, improvements, and discoveries (herein referred to as the "Invention") disclosed in the above-identified patent application and further identified by the IBM Docket Number provided above in the header of this document;

Whereas, International Business Machines Corporation, a corporation of New York having a place of business at Armonk, New York (herein referred to as "IBM"), desires to acquire, and each undersigned inventor desires to grant to IBM, the entire worldwide right, title, and interest in and to the Invention and in and to any and all patent applications and patents directed thereto;

Now, therefore, for good and valuable consideration, the receipt and sufficiency thereof being hereby acknowledged, each undersigned inventor ("ASSIGNOR") hereby sells or has sold, assigns or has assigned, and otherwise transfers or has transferred to IBM (the "ASSIGNEE"), its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the Invention, the above-identified United States patent application, and any and all other patent applications and patents for the Invention which may be applied for or granted therefor in the United States and in all foreign countries and jurisdictions, including all divisions, continuations, reissues, reexaminations, renewals, extensions, counterparts, substitutes, and extensions thereof, and all rights of priority resulting from the filing of such applications and granting of such patents. In addition, each undersigned inventor hereby authorizes and requests the Director of the United States Patent and Trademark Office to issue any United States Patent, and foreign patent authorities to issue any foreign patent, granted for the Invention, to IBM, its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the same to be held and enjoyed by IBM, its successors, legal representatives, and assigns to the full end of the terms for which any and all such patents may be granted, as fully and entirely as would have been held and enjoyed by the undersigned had this Assignment not been made; and each undersigned inventor agrees to execute any and all documents and instruments and perform all lawful acts reasonably related to recording this Assignment or perfecting title to the Invention and all related patents and applications, in IBM, its successors, legal representatives, and assigns, whenever requested by IBM, its successors, legal representatives, or assigns.

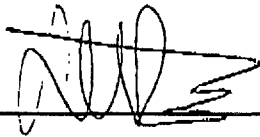
Each undersigned inventor acknowledges their prior and ongoing obligations to sell, assign, and transfer the rights under this Assignment to IBM and is unaware of any reason why they may not have the full and unencumbered right to

sell, assign, and transfer the rights hereby sold, assigned, and transferred, and has not executed, and will not execute, any document or instrument in conflict herewith. Each undersigned inventor also hereby grants IBM, its successors, legal representatives, and assigns, the right to insert in this Assignment any further identification (including, but not limited to, patent Application Number) which may be necessary or desirable for recordation of this Assignment. This Assignment is governed by the substantive laws of the State of New York, and any disputes will be resolved in a New York state court or federal court sited in New York.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

(1) Legal Name of Inventor: **Emre Alptekin**

Signature: _____



Date: 05/19/2015

(2) Legal Name of Inventor: **Nicolas L. Breil**

Signature: _____ Date: _____

(3) Legal Name of Inventor: **Christian Lavoie**

Signature: _____ Date: _____

(4) Legal Name of Inventor: **Ahmet S. Ozcan**

Signature: _____ Date: _____

(5) Legal Name of Inventor: **Kathryn T. Schonenberg**

Signature: _____ Date: _____

**DECLARATION (37 C.F.R. 1.63) FOR UTILITY PATENT APPLICATION USING AN
APPLICATION DATA SHEET (37 C.F.R. 1.76) AND ASSIGNMENT**

Title of Invention: **CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING
SAME**

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☒ United States application or PCT international application number 14/729,298 filed on
June 3, 2015.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I have reviewed and understand the contents of the application, including the claims.

I am aware of the duty to disclose to the United States Patent and Trademark Office all information known to me to be material to patentability as defined in 37 CFR Section 1.56.

Whereas, the undersigned inventor(s) has/have made certain inventions, improvements, and discoveries (herein referred to as the "Invention") disclosed in the above-identified patent application and further identified by the IBM Docket Number provided above in the header of this document;

Whereas, International Business Machines Corporation, a corporation of New York having a place of business at Armonk, New York (herein referred to as "IBM"), desires to acquire, and each undersigned inventor desires to grant to IBM, the entire worldwide right, title, and interest in and to the Invention and in and to any and all patent applications and patents directed thereto;

Now, therefore, for good and valuable consideration, the receipt and sufficiency thereof being hereby acknowledged, each undersigned inventor ("ASSIGNOR") hereby sells or has sold, assigns or has assigned, and otherwise transfers or has transferred to IBM (the "ASSIGNEE"), its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the Invention, the above-identified United States patent application, and any and all other patent applications and patents for the Invention which may be applied for or granted therefor in the United States and in all foreign countries and jurisdictions, including all divisions, continuations, reissues, reexaminations, renewals, extensions, counterparts, substitutes, and extensions thereof, and all rights of priority resulting from the filing of such applications and granting of such patents. In addition, each undersigned inventor hereby authorizes and requests the Director of the United States Patent and Trademark Office to issue any United States Patent, and foreign patent authorities to issue any foreign patent, granted for the Invention, to IBM, its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the same to be held and enjoyed by IBM, its successors, legal representatives, and assigns to the full end of the terms for which any and all such patents may be granted, as fully and entirely as would have been held and enjoyed by the undersigned had this Assignment not been made; and each undersigned inventor agrees to execute any and all documents and instruments and perform all lawful acts reasonably related to recording this Assignment or perfecting title to the Invention and all related patents and applications, in IBM, its successors, legal representatives, and assigns, whenever requested by IBM, its successors, legal representatives, or assigns.

Each undersigned inventor acknowledges their prior and ongoing obligations to sell, assign, and transfer the rights under this Assignment to IBM and is unaware of any reason why they may not have the full and unencumbered right to

sell, assign, and transfer the rights hereby sold, assigned, and transferred, and has not executed, and will not execute, any document or instrument in conflict herewith. Each undersigned inventor also hereby grants IBM, its successors, legal representatives, and assigns, the right to insert in this Assignment any further identification (including, but not limited to, patent Application Number) which may be necessary or desirable for recordation of this Assignment. This Assignment is governed by the substantive laws of the State of New York, and any disputes will be resolved in a New York state court or federal court sited in New York.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

(1) Legal Name of Inventor: **Emre Alptekin**

Signature: _____ Date: _____

(2) Legal Name of Inventor: **Nicolas L. Breil**

Signature: Nicolas L. Breil Date: 5/20/2015

(3) Legal Name of Inventor: **Christian Lavoie**

Signature: _____ Date: _____

(4) Legal Name of Inventor: **Ahmet S. Ozcan**

Signature: _____ Date: _____

(5) Legal Name of Inventor: **Kathryn T. Schonenberg**

Signature: _____ Date: _____

**DECLARATION (37 C.F.R. 1.63) FOR UTILITY PATENT APPLICATION USING AN
APPLICATION DATA SHEET (37 C.F.R. 1.76) AND ASSIGNMENT**

Title of Invention: **CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING
SAME**

As a below named inventor, I hereby declare that:

This declaration is directed to the attached application, or (if following box is checked):

☒ United States application or PCT international application number 14/729,298 filed on
June 3, 2015.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I have reviewed and understand the contents of the application, including the claims.

I am aware of the duty to disclose to the United States Patent and Trademark Office all information known to me to be material to patentability as defined in 37 CFR Section 1.56.

Whereas, the undersigned inventor(s) has/have made certain inventions, improvements, and discoveries (herein referred to as the "Invention") disclosed in the above-identified patent application and further identified by the IBM Docket Number provided above in the header of this document;

Whereas, International Business Machines Corporation, a corporation of New York having a place of business at Armonk, New York (herein referred to as "IBM"), desires to acquire, and each undersigned inventor desires to grant to IBM, the entire worldwide right, title, and interest in and to the Invention and in and to any and all patent applications and patents directed thereto;

Now, therefore, for good and valuable consideration, the receipt and sufficiency thereof being hereby acknowledged, each undersigned inventor ("ASSIGNOR") hereby sells or has sold, assigns or has assigned, and otherwise transfers or has transferred to IBM (the "ASSIGNEE"), its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the Invention, the above-identified United States patent application, and any and all other patent applications and patents for the Invention which may be applied for or granted therefor in the United States and in all foreign countries and jurisdictions, including all divisions, continuations, reissues, reexaminations, renewals, extensions, counterparts, substitutes, and extensions thereof, and all rights of priority resulting from the filing of such applications and granting of such patents. In addition, each undersigned inventor hereby authorizes and requests the Director of the United States Patent and Trademark Office to issue any United States Patent, and foreign patent authorities to issue any foreign patent, granted for the Invention, to IBM, its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the same to be held and enjoyed by IBM, its successors, legal representatives, and assigns to the full end of the terms for which any and all such patents may be granted, as fully and entirely as would have been held and enjoyed by the undersigned had this Assignment not been made; and each undersigned inventor agrees to execute any and all documents and instruments and perform all lawful acts reasonably related to recording this Assignment or perfecting title to the Invention and all related patents and applications, in IBM, its successors, legal representatives, and assigns, whenever requested by IBM, its successors, legal representatives, or assigns.

Each undersigned inventor acknowledges their prior and ongoing obligations to sell, assign, and transfer the rights under this Assignment to IBM and is unaware of any reason why they may not have the full and unencumbered right to

sell, assign, and transfer the rights hereby sold, assigned, and transferred, and has not executed, and will not execute, any document or instrument in conflict herewith. Each undersigned inventor also hereby grants IBM, its successors, legal representatives, and assigns, the right to insert in this Assignment any further identification (including, but not limited to, patent Application Number) which may be necessary or desirable for recordation of this Assignment. This Assignment is governed by the substantive laws of the State of New York, and any disputes will be resolved in a New York state court or federal court sited in New York.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

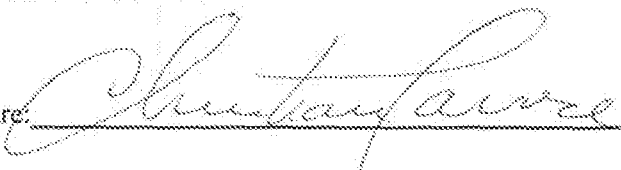
(1) Legal Name of Inventor: **Emre Alptekin**

Signature: _____ Date: _____

(2) Legal Name of Inventor: **Nicolas L. Breil**

Signature: _____ Date: _____

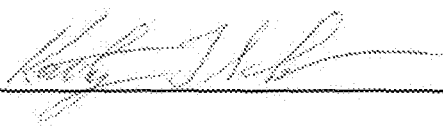
(3) Legal Name of Inventor: **Christian Lavoie**

Signature:  Date: May 13 2015

(4) Legal Name of Inventor: **Ahmet S. Ozcan**

Signature: _____ Date: _____

(5) Legal Name of Inventor: **Kathryn T. Schonenberg**

Signature:  Date: May 13, 2015

**DECLARATION (37 C.F.R. 1.63) FOR UTILITY PATENT APPLICATION USING AN
APPLICATION DATA SHEET (37 C.F.R. 1.76) AND ASSIGNMENT**

Title of Invention: **CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING
SAME**

As a below named inventor, I hereby declare that:

This declaration is directed to the attached application, or (if following box is checked):

☒ United States application or PCT international application number 14/729,298 filed on
June 3, 2015.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I have reviewed and understand the contents of the application, including the claims.

I am aware of the duty to disclose to the United States Patent and Trademark Office all information known to me to be material to patentability as defined in 37 CFR Section 1.56.

Whereas, the undersigned inventor(s) has/have made certain inventions, improvements, and discoveries (herein referred to as the "Invention") disclosed in the above-identified patent application and further identified by the IBM Docket Number provided above in the header of this document;

Whereas, International Business Machines Corporation, a corporation of New York having a place of business at Armonk, New York (herein referred to as "IBM"), desires to acquire, and each undersigned inventor desires to grant to IBM, the entire worldwide right, title, and interest in and to the Invention and in and to any and all patent applications and patents directed thereto;

Now, therefore, for good and valuable consideration, the receipt and sufficiency thereof being hereby acknowledged, each undersigned inventor ("ASSIGNOR") hereby sells or has sold, assigns or has assigned, and otherwise transfers or has transferred to IBM (the "ASSIGNEE"), its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the Invention, the above-identified United States patent application, and any and all other patent applications and patents for the Invention which may be applied for or granted therefor in the United States and in all foreign countries and jurisdictions, including all divisions, continuations, reissues, reexaminations, renewals, extensions, counterparts, substitutes, and extensions thereof, and all rights of priority resulting from the filing of such applications and granting of such patents. In addition, each undersigned inventor hereby authorizes and requests the Director of the United States Patent and Trademark Office to issue any United States Patent, and foreign patent authorities to issue any foreign patent, granted for the Invention, to IBM, its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the same to be held and enjoyed by IBM, its successors, legal representatives, and assigns to the full end of the terms for which any and all such patents may be granted, as fully and entirely as would have been held and enjoyed by the undersigned had this Assignment not been made; and each undersigned inventor agrees to execute any and all documents and instruments and perform all lawful acts reasonably related to recording this Assignment or perfecting title to the Invention and all related patents and applications, in IBM, its successors, legal representatives, and assigns, whenever requested by IBM, its successors, legal representatives, or assigns.

Each undersigned inventor acknowledges their prior and ongoing obligations to sell, assign, and transfer the rights under this Assignment to IBM and is unaware of any reason why they may not have the full and unencumbered right to

sell, assign, and transfer the rights hereby sold, assigned, and transferred, and has not executed, and will not execute, any document or instrument in conflict herewith. Each undersigned inventor also hereby grants IBM, its successors, legal representatives, and assigns, the right to insert in this Assignment any further identification (including, but not limited to, patent Application Number) which may be necessary or desirable for recordation of this Assignment. This Assignment is governed by the substantive laws of the State of New York, and any disputes will be resolved in a New York state court or federal court sited in New York.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

(1) Legal Name of Inventor: **Emre Alptekin**

Signature: _____ Date: _____

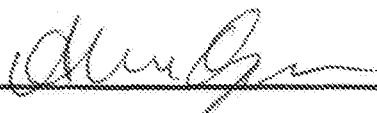
(2) Legal Name of Inventor: **Nicolas L. Breil**

Signature: _____ Date: _____

(3) Legal Name of Inventor: **Christian Lavoie**

Signature: _____ Date: _____

(4) Legal Name of Inventor: **Ahmet S. Ozcan**

Signature:  Date: 5/13/2015

(5) Legal Name of Inventor: **Kathryn T. Schonenberg**

Signature: _____ Date: _____

Electronic Acknowledgement Receipt

EFS ID:	32266361
Application Number:	15826939
International Application Number:	
Confirmation Number:	6013
Title of Invention:	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME
First Named Inventor/Applicant Name:	Emre Alptekin
Customer Number:	133964
Filer:	Justine M. Wilbur
Filer Authorized By:	
Attorney Docket Number:	FIS920150066US2
Receipt Date:	06-APR-2018
Filing Date:	30-NOV-2017
Time Stamp:	14:04:44
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Oath or Declaration filed	FIS920150066US1_Filed_Dec-Assign_04-06-18.pdf	811062	no	8
			0e7dc530b943b919b8307a51775999ec2ef0d613		

Warnings:

Information:	
Total Files Size (in bytes):	811062
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.	



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/826,939	11/30/2017	Emre Alptekin	FIS920150066US2	6013

133964 7590 08/27/2018
HOFFMAN WARNICK LLC
540 Broadway
4th Floor
ALBANY, NY 12207

EXAMINER

NGUYEN, LAWRENCE-LIN T

ART UNIT	PAPER NUMBER
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2897

NOTIFICATION DATE	DELIVERY MODE
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08/27/2018

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

PTOCommunications@hoffmanwarnick.com

Office Action Summary	Application No. 15/826,939	Applicant(s) ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T. NGUYEN	Art Unit 2897	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 11/30/2017.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 1-6 is/are pending in the application.
5a) Of the above claim(s) _____ is/are withdrawn from consideration.
- 6) ☐ Claim(s) _____ is/are allowed.
- 7) ☒ Claim(s) 1-6 is/are rejected.
- 8) ☐ Claim(s) _____ is/are objected to.
- 9) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☐ The drawing(s) filed on _____ is/are: a) ☐ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- 1) ☒ Notice of References Cited (PTO-892)
- 2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date 11/30/2017.
- 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____.
- 4) ☐ Other: _____.

Detailed Action

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.
2. This non-final Office action is responsive to Applicants' application filed **04/06/2018**. Claims **1-6** are presented for examination and are rejected for the reasons indicated herein below.
3. For applicant's benefit portions of the cited reference(s) have been cited to aid in the review of the rejection(s). While every attempt has been made to be thorough and consistent within the rejection it is noted that the *PRIOR ART MUST BE CONSIDERED IN ITS ENTIRETY, INCLUDING DISCLOSURES THAT TEACH AWAY FROM THE CLAIMS*. See MPEP 2141.02 VI.

Claim Rejections - 35 USC § 102(a)(2)

4. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

5. Claims **1-6** are rejected under 35 U.S.C. 102(a2), as being anticipated by **Alptekin et al., (US2015/0221740A1, hereinafter Alptekin)**.

6. In reclaim **1**, in FIG. 1-9, Alptekin discloses a semiconductor structure including a contact(**36**) to a semiconductor substrate(**10**), the contact(**36**) comprising:

- a contact opening(**30S&D**) in a dielectric layer(**26P**) over the semiconductor substrate(**10**);
- a first liner layer(**32L**) substantially coating a sidewall the contact opening(**30S&D**);
- a refractory metal layer(**34L**) substantially coating the first liner layer(**32L**) in the contact opening(**30S&D**);
- a metal(**36**) substantially filling the contact opening(**30S&D**); and
- an intermix region(**38S&D**) at an upper surface of the semiconductor

substrate(10) in the contact opening(30S&D), the intermix(38S&D) region having a composition including a material used in the first liner layer(32L), a material used in the refractory metal layer(34L), and a material used in the semiconductor substrate(10) (See § [0031-0078]).

7. In reclaim 2, in FIG. 1-9, Alptekin discloses a contact of claim 1, wherein the intermix region(38S&D) includes a titanium-based silicide (See § [0065]).

8. In reclaim 3, in FIG. 1-9, Alptekin discloses a contact of claim 1, wherein the first liner layer(32L) includes titanium (Ti) and the refractory metal layer(34L) includes titanium nitride (TiN) (See § [0078]).

9. In reclaim 4, in FIG. 1-9, Alptekin discloses a contact of claim 1, wherein the first liner layer(32L) includes nickel platinum (NiPt) and the refractory metal layer(32L) includes a bilayer having a first refractory metal material including titanium (Ti) and a second refractory metal material including titanium (TiN), the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal(36) (See § [0078]).

10. In reclaim 5, in FIG. 1-9, Alptekin discloses a contact of claim 1, wherein the metal(32L) includes tungsten (W) (See § [0078]).

11. In reclaim 6, in FIG. 9, Alptekin discloses a contact of claim 1, wherein the intermix region(38) has a thickness of approximately 3 nanometers to approximately 5 nanometers (See § [0077]).

Conclusion

13. Any inquiry concerning this communication or earlier communications from the examiner should be directed to LAWRENCE-LINH T. NGUYEN at Lawrence-Linh.Nguyen@uspto.gov, or telephone number (571) 272-6267, or fax number (571) 273-6267. The examiner can be reached on M-F (8AM - 4PM EST).

14. If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Fernando Toledo can be reached on (571) 272-1867. The fax phone number for the organization where this application or proceeding is assigned is (571) 272-8300.

15. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at (866) 217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call (800) 786-9199 (in USA or CANADA) or (571) 272-1000.

/Lawrence-Linh T Nguyen/

Examiner, Art Unit 2897

/Fernando L. Toledo/

Supervisory Patent Examiner, Art
Unit 2897

Notice of References Cited	Application/Control No. 15/826,939		Applicant(s)/Patent Under Reexamination ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T. NGUYEN		Art Unit 2897	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-2015/0221740 A1	08-2015	Alptekin; Emre	H01L29/456	257/288
	B	US-				
	C	US-				
	D	US-				
	E	US-				
	F	US-				
	G	US-				
	H	US-				
	I	US-				
	J	US-				
	K	US-				
	L	US-				
	M	US-				

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

<i>Index of Claims</i> 	Application/Control No. 15826939	Applicant(s)/Patent Under Reexamination ALPTEKIN ET AL.
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	08/22/2018								
	1	✓								
	2	✓								
	3	✓								
	4	✓								
	5	✓								
	6	✓								

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	4	"14729298" "15826939"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/08/21 23:19
L2	1	"15826939"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/08/21 23:21
S1	0	("14729298").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2016/09/12 20:02
S2	48	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:02
S3	367	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:02
S4	104	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:03
S5	133	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:03
S6	104	((Emre) near2 (Alptekin)).INV.	US-PGPUB;	OR	ON	2016/09/12

			USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			20:21
S7	9310	("GLOBALFOUNDRIES INC." GLOBALFOUNDRIES).AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:22
S8	55	("8278200").PN. ("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 21:07
S9	181	"7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:21
S10	26	(US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US- 20130009217-\$ or US-20090289302-\$ or US-20140145319-\$ or US- 20140138837-\$ or US-20090302480-\$ or US-20100078771-\$ or US- 20050085064-\$ or US-20060246656-\$ or US-20120132966-\$) .did.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:22
S11	13120	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:24
S12	23052	S1 S2 S3 S4 S5 S6 S7 S11 S9 S10	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:25
S13	1273	S12 and ((opening trench via hole) with dielectric) and ((liner or barrier) with metal) and (metal with (contact or fill))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT;	OR	ON	2016/09/12 22:25

			IBM_TDB			
S14	3241	S12 and ((Drain Gate Source Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/12 22:25
S15	92	S12 and ((opening or trench or via or hole) with dielectric) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:26
S16	516	S13 S14	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/12 22:27
S17	28	S16 and ((opening or trench or via or hole) with dielectric) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:27
S18	11581	((Drain and Gate and Source and Spacer) with (Trench Open Cavity Contact adj Open Contact Metal ((Metal Contact) adj Liner)) with (Dielectric Isolat\$3 Liner Mask Oxide))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:44
S19	650	S12 and ((Drain and Gate and Source and Spacer) with (Trench Open Cavity Contact adj Open Contact Metal ((Metal Contact) adj Liner)) with (Dielectric Isolat\$3 Liner Mask Oxide))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:45
S20	110	S12 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:47
S21	22	S19 S20	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/12 22:48
S22	170152	((Gate Spacer))	US-PGPUB; USPAT; USOCR; FPRS;	AND	ON	2016/09/12 22:52

			EPO; JPO; DERWENT; IBM_TDB			
S23	63156	S22 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:55
S24	22129	S22 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:58
S25	1183	S22 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:58
S26	2141	S22 and (("W" Tungsten "Ti" Titanium "Ta" Tantalum "Mo" Molybdenum "TiN" Titanium adj Nitride "TaN" Tantalum adj Nitride Titanium-Tungsten Titanium adj Tungsten) with ("W" Tungsten "Co" Colbalt "Cu" Copper and "Al" Aluminum) with ("NiPt" Nickel-Platinum Nickel adj Platinum "Ti/TiN" bilayer Liner/Barrier Liner adj2 Barrier))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 23:09
S27	1	"14729298"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S28	49	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S29	368	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S30	105	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03

S31	134	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S32	105	((Emre) near2 (Alptekin)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S33	245794	("GLOBALFOUNDRI ES INC." GLOBALFOUNDRI ES "IBM" International adj Business adj Machines "International Business Machines").AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:05
S34	13131	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:12
S35	276	("8278200").PN. ("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN. "7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502" (US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US- 20130009217-\$ or US-20090289302-\$ or US-20140145319-\$ or US- 20140138837-\$ or US-20090302480-\$ or US-20100078771-\$ or US- 20050085064-\$ or US-20060246656-\$ or US-20120132966-\$) .did. ("20070059878" "7253071" "8101518" "20070181954" "20070093047" "20020111021" "6346465" "20130119483" "20070134898" "6468901" "20080121950" "20100224938" "20120074501" "20070202692"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:13

		"8338239").PN.				
S36	259027	S27 S28 S29 S30 S31 S32 S33 S34 S35	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:15
S37	10581	S36 and ((Drain Gate Source Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:25
S38	29272	S36 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Insulat\$3 Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:25
S39	261	S36 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:28
S40	356	S36 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:28
S41	6783	S36 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:29
S42	8119	S37 S38	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:29
S43	2155	S37 S38 S41	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:29
S44	6	S39 S40	US-PGPUB; USPAT; USOCR; FPRS;	AND	ON	2016/09/13 13:29

			EPO; JPO; DERWENT; IBM_TDB			
S45	0	S43 S44	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:29
S46	43	S43 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:42
S47	9	S43 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:43
S48	1709	S43 and ((silicon "Si" germanium "Ge" silicon adj germanium "SiGe" "Si.sub" adj "Ge.sub" silicon adj carbide "SiC" "III-V") near3 (Substrate submount wafer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 16:49
S49	78	S43 and ((Gate adj Dielectric Gate) adj3 ("HfO2" "HfO.sub.2" "ZrO2" "ZrO.sub.2" "La2O3" "La.sub.2O.sub.3" "AlzO3" "Al.sub.zO.sub.3" "TiO2" "TiO.sub.2" "SrTiO3" "SrTiO.sub.3" "LaAlO3" "LaAlO.sub.3" "Y2O3" "Y.sub.2O.sub.3" "HfOxNy" "HfO.sub.xN.sub.y" "ZrOxNy" "ZrO.sub.xN.sub.y" "La2OxNy" "La.sub.2O.sub.xN.sub.y" "AlzOxNy" "Al.sub.zO.sub.xN.sub.y" "TiOxNy" "TiO.sub.xN.sub.y" "SrTiOxNy" "SrTiO.sub.xN.sub.y" "LaAlOxNy" "LaAlO.sub.xN.sub.y" "Y2OxNy" "Y.sub.2O.sub.xN.sub.y" "SiON" "SiNx" "SiN.sub.x" Silicate))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:11
S50	43406	S36 and ((chemical adj vapor adj deposition "chemical vapor deposition" (CVD) "low-pressure CVD" (LPCVD) "plasma-enhanced CVD" (PECVD) "semi-atmosphere CVD" (SACVD) "high density plasma CVD" "high density plasma" (HDPCVD) "rapid thermal CVD" (RTCVD) "ultra-high vacuum CVD" (UHVCVD) "limited reaction processing CVD" (LRPCVD) "metalorganic CVD" (MOCVD) ((sputtering ion adj beam electron adj beam laser adj assisted physical adj vapor atomic adj layer) adj (Deposit deposition)) "electron beam deposition" "sputtering deposition" "ion beam deposition" "laser assisted	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:30

		deposition" "thermal oxidation" "thermal nitridation spin-on" "physical vapor deposition" (PVD) "atomic layer deposition" (ALD) "chemical oxidation" "molecular beam epitaxy" (MBE) plat\$3 evaporat\$3) ((low-pressure low adj pressure plasma-enhanced plasma adj enhanced semi-atmosphere semi adj atmosphere high adj density adj plasma rapid adj thermal ultra-high adj vacuum ultra adj high adj vacuum limited adj reaction adj processing metalorganic) adj (CVD)))				
S51	1853	S43 and ((chemical adj vapor adj deposition "chemical vapor deposition" (CVD) "low-pressure CVD" (LPCVD) "plasma-enhanced CVD" (PECVD) "semi-atmosphere CVD" (SACVD) "high density plasma CVD" "high density plasma" (HDPCVD) "rapid thermal CVD" (RTCVD) "ultra-high vacuum CVD" (UHVCVD) "limited reaction processing CVD" (LRPCVD) "metalorganic CVD" (MOCVD) ((sputtering ion adj beam electron adj beam laser adj assisted physical adj vapor atomic adj layer) adj (Deposit deposition)) "electron beam deposition" "sputtering deposition" "ion beam deposition" "laser assisted deposition" "thermal oxidation" "thermal nitridation spin-on" "physical vapor deposition" (PVD) "atomic layer deposition" (ALD) "chemical oxidation" "molecular beam epitaxy" (MBE) plat\$3 evaporat\$3) ((low-pressure low adj pressure plasma-enhanced plasma adj enhanced semi-atmosphere semi adj atmosphere high adj density adj plasma rapid adj thermal ultra-high adj vacuum ultra adj high adj vacuum limited adj reaction adj processing metalorganic) adj (CVD)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:31
S52	1481	S48 S51	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 17:32
S53	62	S52 and ((Gate adj Dielectric Gate) adj3 ("HfO2" "HfO.sub.2" "ZrO2" "ZrO.sub.2" "La2O3" "La.sub.2O.sub.3" "AlzO3" "Al.sub.zO.sub.3" "TiO2" "TiO.sub.2" "SrTiO3" "SrTiO.sub.3" "LaAlO3" "LaAlO.sub.3" "Y2O3" "Y.sub.2O.sub.3" "HfOxNy" "HfO.sub.xN.sub.y" "ZrOxNy" "ZrO.sub.xN.sub.y" "La2OxNy" "La.sub.2O.sub.xN.sub.y" "AlzOxNy" "Al.sub.zO.sub.xN.sub.y" "TiOxNy" "TiO.sub.xN.sub.y" "SrTiOxNy" "SrTiO.sub.xN.sub.y" "LaAlOxNy" "LaAlO.sub.xN.sub.y" "Y2OxNy" "Y.sub.2O.sub.xN.sub.y" "SiON" "SiNx"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:32

		"S1N.sub.x" Silicate))				
S54	5	S52 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:32
S55	29	S52 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:32
S57	1481	S52 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:39
S58	1452	S52 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:39
S59	318	S52 and ((Drain and Gate and Source and Spacer) with (Trench Open Cavity Contact adj Open Contact Metal ((Metal Contact) adj Liner)) with (Dielectric Isolat\$3 Liner Mask Oxide))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:40
S60	318	S57 S58 S59	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 17:40
S61	16	("20010035558" "20050124128" "20080171411" "20080272357" "20080303068" "20100075474" "6087209" "6093243" "6440828" "6812082" "7259110" "7736964" "7755114" "7785947").PN. OR ("8304819").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:07
S62	21	("20040056281" "20040121590" "20050020093" "20050026380" "20060141712" "20060166494" "20070042580" "20070141798" "20090134497" "20110084320" "20120202326" "20120248550" "20140015104" "6323519" "6730570" "7087515" "7220668" "7348283" "7679192" "8304819" "8492217").PN. OR ("9252019").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:15
S63	2	"20130049219"	US-PGPUB;	OR	ON	2016/09/14

			USPAT; USOCR			14:43
S64	21	("20040056281" "20040121590" "20050020093" "20050026380" "20060141712" "20060166494" "20070042580" "20070141798" "20090134497" "20110084320" "20120202326" "20120248550" "20140015104" "6323519" "6730570" "7087515" "7220668" "7348283" "7679192" "8304819" "8492217").PN. OR ("9252019").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:44
S65	13	("20040067658" "20040150075" "20050110153" "20060003577" "6878620" "7087515" "7199044" "7253097" "7253524" "7309658" "7466027").PN. OR ("7679192").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:49
S66	4	"20130049219" "20070152334"	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:51
S67	1	"14729298"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S68	49	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S69	368	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S70	105	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S71	134	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S72	105	((Emre) near2 (Alptekin)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT;	OR	ON	2016/09/14 15:09

			IBM_TDB			
S73	245794	("GLOBALFOUNDRIES INC." GLOBALFOUNDRIES "IBM" International adj Business adj Machines "International Business Machines").AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S74	13131	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S75	276	("8278200").PN. ("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN. "7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502" (US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US- 20130009217-\$ or US-20090289302-\$ or US-20140145319-\$ or US- 20140138837-\$ or US-20090302480-\$ or US-20100078771-\$ or US- 20050085064-\$ or US-20060246656-\$ or US-20120132966-\$) .did. ("20070059878" "7253071" "8101518" "20070181954" "20070093047" "20020111021" "6346465" "20130119483" "20070134898" "6468901" "20080121950" "20100224938" "20120074501" "20070202692" "8338239").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S76	259027	S67 S68 S69 S70 S71 S72 S73 S74 S75	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S77	10581	S76 and ((Drain Gate Source Spacer))	US-PGPUB; USPAT; USOCR; FPRS;	AND	ON	2016/09/14 15:09

			EPO; JPO; DERWENT; IBM_TDB			
S78	29272	S76 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Insulat\$3 Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S79	6783	S76 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S80	2155	S77 S78 S79	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/14 15:09
S81	1709	S80 and ((silicon "Si" germanium "Ge" silicon adj germanium "SiGe" "Si.sub" adj "Ge.sub" silicon adj carbide "SiC" "III-V") near3 (Substrate submount wafer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S82	1853	S80 and ((chemical adj vapor adj deposition "chemical vapor deposition" (CVD) "low-pressure CVD" (LPCVD) "plasma-enhanced CVD" (PECVD) "semi-atmosphere CVD" (SACVD) "high density plasma CVD" "high density plasma" (HDPCVD) "rapid thermal CVD" (RTCVD) "ultra-high vacuum CVD" (UHVCVD) "limited reaction processing CVD" (LRPCVD) "metalorganic CVD" (MOCVD) ((sputtering ion adj beam electron adj beam laser adj assisted physical adj vapor atomic adj layer) adj (Deposit deposition)) "electron beam deposition" "sputtering deposition" "ion beam deposition" "laser assisted deposition" "thermal oxidation" "thermal nitridation spin-on" "physical vapor deposition" (PVD) "atomic layer deposition" (ALD) "chemical oxidation" "molecular beam epitaxy" (MBE) plat\$3 evaporat\$3) ((low-pressure low adj pressure plasma-enhanced plasma adj enhanced semi-atmosphere semi adj atmosphere high adj density adj plasma rapid adj thermal ultra-high adj vacuum ultra adj high adj vacuum limited adj reaction adj processing metalorganic) adj (CVD)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S83	1481	S81 S82	US-PGPUB; USPAT;	AND	ON	2016/09/14 15:09

			USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			
S84	34	S83 and (("W" Tungsten "Ti" Titanium "Ta" Tantalum "Mo" Molybdenum "TiN" Titanium adj Nitride "TaN" Tantalum adj Nitride Titanium-Tungsten Titanium adj Tungsten) with ("W" Tungsten "Co" Colbalt "Cu" Copper and "Al" Aluminum) with ("NiPt" Nickel-Platinum Nickel adj Platinum "Ti/TiN" bilayer Liner/Barrier Liner adj2 Barrier))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S85	8119	S77 S78	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/14 17:08
S86	2155	S77 S78 S79	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/14 17:08
S87	407	S76 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) with (Remov\$3 Extract\$3 Etch\$3) with (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 17:35
S88	45	S86 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) with (Remov\$3 Extract\$3 Etch\$3) with (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 17:37
S89	0	S86 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) adj5 (Remov\$3 Extract\$3 Etch\$3) adj5 (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 20:05
S90	118	S86 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) adj5 (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 20:06
S91	33	S90 and ((Remov\$3 Extract\$3 Etch\$3) adj5 (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2016/09/14 20:06

			DERWENT; IBM_TDB			
S92	1480	S80 and (chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP) "laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 21:52
S93	22	(US-20150221740-\$ or US-20130240990-\$ or US-20100078771-\$ or US-20090023261-\$ or US-20120161324-\$ or US-20130119483-\$ or US-20140319620-\$ or US-20140246698-\$ or US-20120104498-\$ or US-20080040697-\$ or US-20060163631-\$ or US-20070122956-\$ or US-20130049219-\$ or US-20070152334-\$).did. or (US-8304819-\$ or US-9224675-\$ or US-8742510-\$ or US-9093425-\$ or US-9252019-\$ or US-6440828-\$ or US-7679192-\$).did. or (US-20110084320-\$).did.	US-PGPUB; USPAT; DERWENT	OR	ON	2016/09/14 21:52
S94	16	S93 and (chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP) "laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 21:52
S95	4	S93 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) and ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 21:57
S96	22455	H01L21/7684 H01L21/76841 H01L21/76843 H01L21/76846 H01L21/7685 H01L21/76879 H01L21/76883 H01L23/481 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 10:32
S97	69	"20150187896" "5565708"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 10:33
S98	639	S96 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) and ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:44
S99	21	S98 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric	US-PGPUB; USPAT; USOCR;	OR	ON	2016/09/15 12:44

		Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	FPRS; EPO; JPO; DERWENT; IBM_TDB			
S100	1	"14729298"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S101	49	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S102	368	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S103	105	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S104	134	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S105	105	((Emre) near2 (Alptekin)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S106	245815	("GLOBALFOUNDRIES INC." GLOBALFOUNDRIES "IBM" International adj Business adj Machines "International Business Machines").AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S107	13151	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S108	276	("8278200").PN. ("20070249149"	US-PGPUB;	OR	ON	2016/09/15

		"20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN. "7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502" (US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US- 20130009217-\$ or US-20090289302-\$ or US-20140145319-\$ or US- 20140138837-\$ or US-20090302480-\$ or US-20100078771-\$ or US- 20050085064-\$ or US-20060246656-\$ or US-20120132966-\$) .did. ("20070059878" "7253071" "8101518" "20070181954" "20070093047" "20020111021" "6346465" "20130119483" "20070134898" "6468901" "20080121950" "20100224938" "20120074501" "20070202692" "8338239").PN.	USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			12:45
S109	259067	S100 S101 S102 S103 S104 S105 S106 S107 S108	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S110	19	S109 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) adj5 (Remov\$3 Extract\$3 Etch\$3) adj5 (Source Drain)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:57
S111	20386	S109 and (chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP) "laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 13:52
S112	4016	S109 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) and ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 13:53

S113	0	S98 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" ("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) adj5 (Remov\$3 Extract\$3 Etch\$3) adj5 (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 13:54
S114	0	S112 and ((First Second Top Bottom Upper Lower High\$3 Low\$3 Left Right Adjacent) adj (refractory adj (material metal)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:00
S115	84	S109 and ((First Second Top Bottom Upper Lower High\$3 Low\$3 Left Right Adjacent) adj (refractory adj (material metal)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:01
S116	0	S98 and ((First Second Top Bottom Upper Lower High\$3 Low\$3 Left Right Adjacent) adj (refractory adj (material metal)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:01
S117	34	("20020019107" "20020102810" "20040206620" "20040259308" "20050087879" "20050263890" "20070145525" "20080073708" "20080138948" "20080188055" "20080272421" "20100001325" "20100025853" "20100187655" "20100213571" "20110057292" "20130011990" "5981331" "6219125" "6511879" "6593616" "6746916" "6753618" "6881642" "6906420" "7071054" "7268072" "7304367" "7329939" "7368343" "7545009" "7772678" "8105891").PN. OR ("8835274").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/15 14:04
S118	2	"20140027883"	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/15 14:04
S119	0	S115 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) and ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:11
S120	0	S115 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) with ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:12
S121	36	S115 and ((chemical adj mechanical adj	US-PGPUB;	OR	ON	2016/09/15

		polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			14:12
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8/ 22/ 2018 12:48:12 AM
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Doc code: IDS

PTO/SB/08a (03-15)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 07/31/2016. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number

Filing Date

First Named Inventor

Emre Alptekin

Art Unit

Examiner Name

Attorney Docket Number

FIS920150066US2

U.S.PATENTS

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Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	8278200	B2	2012-10-02	Lavoie et al.	

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Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	20120161324	A1	2012-06-28	Heinrich et al.	
	2	20140027883	A1	2014-01-30	Hu	
	3	20150220791	A1	2015-08-06	Wu et al.	

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	1							

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	Emre Alptekin	
Art Unit		
Examiner Name		
Attorney Docket Number	FIS920150066US2	

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NON-PATENT LITERATURE DOCUMENTS

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1		

If you wish to add additional non-patent literature document citation information please click the Add button

EXAMINER SIGNATURE

Examiner Signature	/LAWRENCE-LIN T NGUYEN/	Date Considered	08/21/2018
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	Emre Alptekin	
Art Unit		
Examiner Name		
Attorney Docket Number	FIS920150066US2	

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Justine M. Wilbur/	Date (YYYY-MM-DD)	2017-11-30
Name/Print	Justine M. Wilbur	Registration Number	59,678

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8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
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<i>Search Notes</i>	Application/Control No.	Applicant(s)/Patent Under Reexamination
	Examiner	Art Unit

CPC- SEARCHED		
Symbol	Date	Examiner

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

SEARCH NOTES		
Search Notes	Date	Examiner

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 15/826,939	Filing Date 11/30/2017	☐ To be Mailed
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 ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED – PART I

FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED – PART II

AMENDMENT	(Column 1)	(Column 2)	(Column 3)	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	11/27/2018	CLAIMS REMAINING AFTER AMENDMENT					
	Total (37 CFR 1.16(i))	* 6	Minus	** 20	= 0	x \$100 =	0
	Independent (37 CFR 1.16(h))	* 1	Minus	*** 3	= 0	x \$460 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
TOTAL ADD'L FEE						0	

AMENDMENT	(Column 1)	(Column 2)	(Column 3)	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
		CLAIMS REMAINING AFTER AMENDMENT					
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$ =	
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$ =	
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
TOTAL ADD'L FEE							

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor: Emre Alptekin

Conf. No.: 6013

Serial No.: 15/826,939

Art Unit: 2897

Filed: November 30, 2017

Examiner: Nguyen, Lawrence-Lin T.

Docket No.: FIS920150066US2
(GFFT-0376-DIV)

Title: **CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME**

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT

This paper is being filed in response to the Office Action dated August 27, 2018. Please amend the above-referenced patent application as follows:

AMENDMENTS TO THE CLAIMS

The following listing of the claims replaces all prior versions, and listings, of claims in the application:

1. (Currently Amended) A semiconductor structure including a contact to a semiconductor substrate, the contact comprising:

- a contact opening in a dielectric layer over the semiconductor substrate;
- a first liner layer substantially coating a sidewall of the contact opening;
- a refractory metal layer substantially coating the first liner layer in the contact opening;
- a metal substantially filling the contact opening; and

an intermix region at an upper surface of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate;

wherein the refractory metal layer includes a multi-layer having at least a first refractory metal material and a second refractory metal material, the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal.

2. (Original) The contact of claim 1, wherein the intermix region includes a titanium-based silicide.

3. (Original) The contact of claim 1, wherein the first liner layer includes titanium (Ti) and the refractory metal layer includes titanium nitride (TiN).

4. (Currently Amended) The contact of claim 1, wherein the first liner layer includes nickel platinum (NiPt)₂ ~~and the refractory metal layer includes a bilayer having a~~ the first refractory metal material ~~including~~ includes titanium (Ti) and ~~[[a]]~~ the second refractory metal material ~~including~~ includes titanium nitride (TiN); ~~the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal.~~

5. (Original) The contact of claim 1, wherein the metal includes tungsten (W).

6. (Original) The contact of claim 1, wherein the intermix region has a thickness of approximately 3 nanometers to approximately 5 nanometers.

REMARKS

Claims Status

Claims 1-6 are pending in this application. Claim 1 is currently amended to include subject matter from claim 4, and finds support in, at least, Figure 9. Claim 4 is currently amended for grammatical purposes and to remove redundant language in light of the amendment to claim 1. Claims 2, 3, 5 and 6 remain as originally presented. No new matter is believed to have been entered.

§ 102(a)(2) Rejection

Claims 1-6 are rejected under 35 U.S.C. 102(a)(2) as allegedly being anticipated by Alptekin (US 2015/0221740A1). Applicant respectfully traverses this rejection as follows.

Independent claim 1 recites, in part, “wherein the refractory metal layer includes a multi-layer having at least a first refractory metal material and a second refractory metal material, the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal.”

Alptekin does not disclose second transition metal liner 34L/34p (being equated to Applicant’s refractory metal layer) as having multiple layers as required by Applicant’s claim 1. Alptekin also does not disclose a first material of liner 34L/34p as substantially separating the first transition metal liner 32L/32p (being equated to Applicant’s first liner layer) from a second material of liner 34L/34p as required by Applicant’s claim 1. Moreover, Alptekin does not disclose a second material of liner 34L/34p as substantially separating a first material of liner 34L/34p from the metal of contact 36 as required by Applicant’s claim 1. While Alptekin may

disclose that second transition metal liner 34L/34p may comprise multiple transition metals ([0067]-[0068]), Alptekin does not disclose liner 34L/34p as having multiple layers where each layer separates specified elements. Accordingly, Applicant submits that the Office has misapplied the multiple transition metals of Alptekin's liner 34L/34p to Applicant's multiple layers (*see* rejection of claim 4 in Office Action).

For any one of the above-noted deficiencies of Alptekin and/or the Office's misapplication of Alptekin's teachings to Applicant's claims, Applicant submits that Alptekin fails to disclose or suggest Applicant's claim 1.

Specifically regarding Applicant's claim 3, wherein the first liner layer includes Ti and the refractory metal layer includes TiN, Applicant submits that the Office has again misapplied the teachings of Alptekin. Alptekin states, numerous times, that the second transition metal liner 34L/34p comprises at least one second transition metal element which differs from the at least one first transition metal element (*see e.g.*, [0067]-[0068]). As such, Alptekin cannot read on Applicant's claim 3 since the TiN of the refractory metal layer does not comprise at least one transition metal element which differs from the Ti of the first liner layer (noting that N is not a transition metal element). Accordingly, Applicant submits that Alptekin fails to disclose or suggest Applicant's claim 3 for at least this additional reason.

For reasons similar to those just stated with respect to Applicant's claim 3, Alptekin cannot read on Applicant's claim 4 which requires the first liner layer to include NiPt and the refractory metal layer to include Ti/TiN. Accordingly, Applicant submits that Alptekin fails to disclose or suggest Applicant's claim 4 as well.

In light of the foregoing, Applicant respectfully requests withdrawal of the anticipation rejection over Alptekin.

Also in light of the foregoing, and considering that independent claim 1 has merely been amended to include subject matter from original claim 4, Applicant requests that a next Office Action, should one be issued, be non-final since Applicant's amendments to the claims would not have necessitated a new ground of rejection.

CONCLUSION

Applicant respectfully submits that the Application as presented is in condition for allowance. Should the Examiner believe that anything further is necessary in order to place the application in better condition for allowance, the Examiner is requested to contact the Applicant's undersigned representative at the telephone number listed below.

Respectfully submitted,

/Justine M. Wilbur/
Justine M. Wilbur
Reg. No. 59,678

Date: November 27, 2018

Hoffman Warnick LLC
540 Broadway, 4th Floor
Albany, New York 12207
Phone: (518) 449-0044
Fax: (518) 449-0047

Electronic Acknowledgement Receipt

EFS ID:	34412720
Application Number:	15826939
International Application Number:	
Confirmation Number:	6013
Title of Invention:	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME
First Named Inventor/Applicant Name:	Emre Alptekin
Customer Number:	133964
Filer:	Justine M. Wilbur
Filer Authorized By:	
Attorney Docket Number:	FIS920150066US2
Receipt Date:	27-NOV-2018
Filing Date:	30-NOV-2017
Time Stamp:	18:20:14
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		FIS920150066US2_Amendmen tToOfficeAction1_27NOV2018_ .pdf	109784 f45c75d7a9e74ab247c3428c4d608b8185d e20f9	yes	6

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Amendment/Req. Reconsideration-After Non-Final Reject	1	1
	Claims	2	3
	Applicant Arguments/Remarks Made in an Amendment	4	6

Warnings:

Information:

Total Files Size (in bytes):	109784
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

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NOTICE OF ALLOWANCE AND FEE(S) DUE

133964 7590 03/23/2020
HOFFMAN WARNICK LLC
540 BROADWAY
4TH FLOOR
ALBANY, NY 12207

EXAMINER

TOLEDO, FERNANDO L

ART UNIT

PAPER NUMBER

2897

DATE MAILED: 03/23/2020

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/826,939	11/30/2017	Emre Alptekin	FIS920150066US2	6013

TITLE OF INVENTION: CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0.00	\$0.00	\$1000	06/23/2020

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

133964 7590 03/23/2020
HOFFMAN WARNICK LLC
540 BROADWAY
4TH FLOOR
ALBANY, NY 12207

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/826.939	11/30/2017	Emre Alptekin	FIS920150066US2	6013

TITLE OF INVENTION: CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0.00	\$0.00	\$1000	06/23/2020

EXAMINER	ART UNIT	CLASS-SUBCLASS
TOLEDO, FERNANDO L	2897	257-757000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/826,939	11/30/2017	Emre Alptekin	FIS920150066US2	6013
133964	7590	03/23/2020	EXAMINER	
HOFFMAN WARNICK LLC			TOLEDO, FERNANDO L	
540 BROADWAY			ART UNIT	
4TH FLOOR			PAPER NUMBER	
ALBANY, NY 12207			2897	
DATE MAILED: 03/23/2020				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 15/826,939	Applicant(s) Alptekin et al.	
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 11/27/2018.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-6. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some *c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.
2. ☐ Certified copies of the priority documents have been received in Application No. _____.
3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☒ Notice of References Cited (PTO-892) 2. ☐ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date _____. 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____. 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date _____.	5. ☐ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
---	--

/LAWRENCE-LIN T NGUYEN/ Examiner, Art Unit 2897	/FERNANDO L TOLEDO/ Supervisory Patent Examiner, Art Unit 2897
--	---

DETAIL ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Allowable Subject Matter

2. The following is an examiner's statement of reasons for allowance

a. With respect to independent claim **1**, the prior art, Alptekin et al., (US2015/0221740A1) teaches:

- 1. (Currently Amended) A semiconductor structure including a contact to a semiconductor substrate, the contact comprising: a contact opening in a dielectric layer over the semiconductor substrate; a first liner layer substantially coating a sidewall of the contact opening; a refractory metal layer substantially coating the first liner layer in the contact opening; a metal substantially filling the contact opening; and an intermix region at an upper surface of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate.
- But the prior arts of record do not teach or fairly suggest a limitation of: wherein the refractory metal layer includes a multi-layer having at least a first refractory metal material and a second refractory metal material, the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal

material substantially separating the first refractory metal material and the metal.

- *Since Unreacted (non-diffused) portions of the first liner layer may be removed prior to completion of fabricating the contact. Therefore, aspects of the present invention do not increase the deposited metal thickness in the contact openings. Thus, contact opening fill issues are avoided and overall contact resistance is improved.* There is no reason apparent to the examiner that all the features claimed in claim **1** would be obvious over the prior art.

- Dependent claims **2-6** are respectively dependent on currently amended allowable independent claim **1**. Therefore, claims **2-6** incorporate the allowable limitations of claim **1**. Consequently, claims **2-6** are allowable.

3. Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance".

Conclusion

4. Any inquiry concerning this communication or earlier communications from the examiner should be directed to Lawrence-Linh Nguyen (Email @ Lawrence-Linh.Nguyen@uspto.gov) whose telephone number is (571) 272-6267 and fax number is (571) 273-6267. The examiner can normally be reached on Monday-Friday (Flex hours 10A-6P). If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Fernando Toledo can be reached on (571) 272-1867. The fax phone number for the organization where this application or proceeding is assigned is (571) 273-8300.

5. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/LAWRENCE-LINH T NGUYEN/

Examiner, Art Unit 2897

/FERNANDO L TOLEDO/

Supervisory Patent Examiner, Art Unit 2897

<i>Notice of References Cited</i>	Application/Control No. 15/826,939		Applicant(s)/Patent Under Reexamination Alptekin et al.	
	Examiner LAWRENCE-LINH T NGUYEN		Art Unit 2897	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20150221740-A1	08-2015	Alptekin; Emre	H01L29/456	257/288
	B					
	C					
	D					
	E					
	F					
	G					
	H					
	I					
	J					
	K					
	L					
	M					

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

<i>Search Notes</i> 	Application/Control No. 15/826,939	Applicant(s)/Patent Under Reexamination Alptekin et al.
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897

CPC - Searched*		
Symbol	Date	Examiner

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

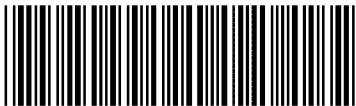
US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Each of the listed CPC, inventor(s) and assignees are searched in combination and/or delimited with text. See EAST search for details.	12/23/2018	

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	12/23/2018	

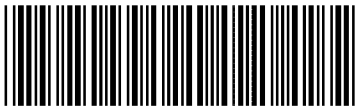
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Issue Classification 	Application/Control No. 15/826,939	Applicant(s)/Patent Under Reexamination Alptekin et al.
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897

CPC						
Symbol					Type	Version
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H01L	/	21	/	76846	I	2013-01-01
H01L	/	21	/	76879	I	2013-01-01
H01L	/	21	/	7684	I	2013-01-01
H01L	/	21	/	76883	I	2013-01-01
H01L	/	23	/	485	I	2013-01-01
H01L	/	21	/	28518	I	2013-01-01
H01L	/	21	/	76855	I	2013-01-01
H01L	/	23	/	53223	A	2013-01-01
H01L	/	23	/	53238	A	2013-01-01
H01L	/	21	/	2855	A	2013-01-01

CPC Combination Sets					
Symbol				Type	Set
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/LAWRENCE-LIN T NGUYEN/ Examiner, Art Unit 2897 (Assistant Examiner)	25 December 2018 (Date)	Total Claims Allowed: 6	
/FERNANDO L TOLEDO/ SPE, Art Unit 2897 (Primary Examiner)	26 December 2018 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 9

Issue Classification 	Application/Control No. 15/826,939	Applicant(s)/Patent Under Reexamination Alptekin et al.
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897

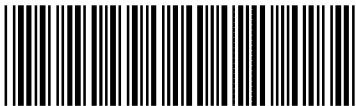
INTERNATIONAL CLASSIFICATION			
CLAIMED			
H01L	/	23	/ 532
H01L	/	21	/ 768

NON-CLAIMED			
	/		/

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

/LAWRENCE-LIN T NGUYEN/ Examiner, Art Unit 2897 (Assistant Examiner)	25 December 2018 (Date)	Total Claims Allowed: 6	
/FERNANDO L TOLEDO/ SPE, Art Unit 2897 (Primary Examiner)	26 December 2018 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 9

Issue Classification 	Application/Control No. 15/826,939	Applicant(s)/Patent Under Reexamination Alptekin et al.
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
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	2														
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/LAWRENCE-LIN T NGUYEN/ Examiner, Art Unit 2897 (Assistant Examiner)	25 December 2018 (Date)	Total Claims Allowed: 6	
/FERNANDO L TOLEDO/ SPE, Art Unit 2897 (Primary Examiner)	26 December 2018 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 9

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
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		"14729298" "15826939"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
		S98 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
		S109 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) adj5 (Remov\$3 Extract\$3 Etch\$3) adj5 (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
		((Emre) near2 (Alptekin)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
		S96 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) and ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
		S93 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) and ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
		((Ahmet) near2 (Ozcan)).INV.	US-PGPUB;	OR	ON	2018/12/10

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		S115 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
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		S112 and ((First Second Top Bottom	US-PGPUB;	OR	ON	2018/12/10

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		S93 and (chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP) "laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
		((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
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			IBM_TDB			
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		S77 S78 S79	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2018/12/10 01:16
		S86 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) adj5 (Remov\$3 Extract\$3 Etch\$3) adj5 (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
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		S76 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) with (Remov\$3 Extract\$3 Etch\$3) with (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
		S81 S82	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2018/12/10 01:16
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		S90 and ((Remov\$3 Extract\$3 Etch\$3) adj5 (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
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S3	367	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:02
S4	104	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:03

S5	133	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:03
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S7	9310	("GLOBALFOUNDRIES INC." GLOBALFOUNDRIES).AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:22
S8	55	("8278200").PN. ("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 21:07
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S12	23052	S1 S2 S3 S4 S5 S6 S7 S11 S9 S10	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2016/09/12 22:25

			DERWENT; IBM_TDB			
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S14	3241	S12 and ((Drain Gate Source Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/12 22:25
S15	92	S12 and ((opening or trench or via or hole) with dielectric) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:26
S16	516	S13 S14	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/12 22:27
S17	28	S16 and ((opening or trench or via or hole) with dielectric) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:27
S18	11581	((Drain and Gate and Source and Spacer) with (Trench Open Cavity Contact adj Open Contact Metal ((Metal Contact) adj Liner)) with (Dielectric Isolat\$3 Liner Mask Oxide))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:44
S19	650	S12 and ((Drain and Gate and Source and Spacer) with (Trench Open Cavity Contact adj Open Contact Metal ((Metal Contact) adj Liner)) with (Dielectric Isolat\$3 Liner Mask Oxide))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:45
S20	110	S12 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:47
S21	22	S19 S20	US-PGPUB; USPAT; USOCR;	AND	ON	2016/09/12 22:48

			FPRS; EPO; JPO; DERWENT; IBM_TDB			
S22	170152	((Gate Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/12 22:52
S23	63156	S22 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:55
S24	22129	S22 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:58
S25	1183	S22 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:58
S26	2141	S22 and (("W" Tungsten "Ti" Titanium "Ta" Tantalum "Mo" Molybdenum "TiN" Titanium adj Nitride "TaN" Tantalum adj Nitride Titanium-Tungsten Titanium adj Tungsten) with ("W" Tungsten "Co" Colbalt "Cu" Copper and "Al" Aluminum) with ("NiPt" Nickel-Platinum Nickel adj Platinum "Ti/TiN" bilayer Liner/Barrier Liner adj2 Barrier))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 23:09
S27	1	"14729298"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S28	49	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S29	368	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT;	OR	ON	2016/09/13 11:03

			IBM_TDB			
S30	105	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S31	134	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S32	105	((Emre) near2 (Alptekin)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S33	245794	("GLOBALFOUNDRIES INC." GLOBALFOUNDRIES "IBM" International adj Business adj Machines "International Business Machines").AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:05
S34	13131	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:12
S35	276	("8278200").PN. ("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN. "7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502" (US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US- 20130009217-\$ or US-20090289302-\$ or US-20140145319-\$ or US- 20140138837-\$ or US-20090302480-\$ or US-20100078771-\$ or US-	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:13

		20050085064-\$ or US-20060246656-\$ or US-20120132966-\$) .did. ("20070059878" "7253071" "8101518" "20070181954" "20070093047" "20020111021" "6346465" "20130119483" "20070134898" "6468901" "20080121950" "20100224938" "20120074501" "20070202692" "8338239").PN.				
S36	259027	S27 S28 S29 S30 S31 S32 S33 S34 S35	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:15
S37	10581	S36 and ((Drain Gate Source Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:25
S38	29272	S36 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Insulat\$3 Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:25
S39	261	S36 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:28
S40	356	S36 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:28
S41	6783	S36 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:29
S42	8119	S37 S38	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:29
S43	2155	S37 S38 S41	US-PGPUB; USPAT;	AND	ON	2016/09/13 13:29

			USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			
S44	6	S39 S40	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:29
S45	0	S43 S44	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:29
S46	43	S43 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:42
S47	9	S43 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:43
S48	1709	S43 and ((silicon "Si" germanium "Ge" silicon adj germanium "SiGe" "Si.sub" adj "Ge.sub" silicon adj carbide "SiC" "III-V") near3 (Substrate submount wafer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 16:49
S49	78	S43 and ((Gate adj Dielectric Gate) adj3 ("HfO2" "HfO.sub.2" "ZrO2" "ZrO.sub.2" "La2O3" "La.sub.2O.sub.3" "AlzO3" "Al.sub.zO.sub.3" "TiO2" "TiO.sub.2" "SrTiO3" "SrTiO.sub.3" "LaAlO3" "LaAlO.sub.3" "Y2O3" "Y.sub.2O.sub.3" "HfOxNy" "HfO.sub.xN.sub.y" "ZrOxNy" "ZrO.sub.xN.sub.y" "La2OxNy" "La.sub.2O.sub.xN.sub.y" "AlzOxNy" "Al.sub.zO.sub.xN.sub.y" "TiOxNy" "TiO.sub.xN.sub.y" "SrTiOxNy" "SrTiO.sub.xN.sub.y" "LaAlOxNy" "LaAlO.sub.xN.sub.y" "Y2OxNy" "Y.sub.2O.sub.xN.sub.y" "SiON" "SiNx" "SiN.sub.x" Silicate))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:11
S50	43406	S36 and ((chemical adj vapor adj deposition "chemical vapor deposition" (CVD) "low-pressure CVD" (LPCVD) "plasma-enhanced CVD" (PECVD) "semi-atmosphere CVD" (SACVD) "high density plasma CVD" "high density plasma" (HDPCVD) "rapid thermal CVD"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:30

		(RTCVD) "ultra-high vacuum CVD" (UHVCVD) "limited reaction processing CVD" (LRPCVD) "metalorganic CVD" (MOCVD) ((sputtering ion adj beam electron adj beam laser adj assisted physical adj vapor atomic adj layer) adj (Deposit deposition)) "electron beam deposition" "sputtering deposition" "ion beam deposition" "laser assisted deposition" "thermal oxidation" "thermal nitridation spin-on" "physical vapor deposition" (PVD) "atomic layer deposition" (ALD) "chemical oxidation" "molecular beam epitaxy" (MBE) plat\$3 evaporat\$3) ((low-pressure low adj pressure plasma-enhanced plasma adj enhanced semi-atmosphere semi adj atmosphere high adj density adj plasma rapid adj thermal ultra-high adj vacuum ultra adj high adj vacuum limited adj reaction adj processing metalorganic) adj (CVD)))				
S51	1853	S43 and ((chemical adj vapor adj deposition "chemical vapor deposition" (CVD) "low-pressure CVD" (LPCVD) "plasma-enhanced CVD" (PECVD) "semi-atmosphere CVD" (SACVD) "high density plasma CVD" "high density plasma" (HDPCVD) "rapid thermal CVD" (RTCVD) "ultra-high vacuum CVD" (UHVCVD) "limited reaction processing CVD" (LRPCVD) "metalorganic CVD" (MOCVD) ((sputtering ion adj beam electron adj beam laser adj assisted physical adj vapor atomic adj layer) adj (Deposit deposition)) "electron beam deposition" "sputtering deposition" "ion beam deposition" "laser assisted deposition" "thermal oxidation" "thermal nitridation spin-on" "physical vapor deposition" (PVD) "atomic layer deposition" (ALD) "chemical oxidation" "molecular beam epitaxy" (MBE) plat\$3 evaporat\$3) ((low-pressure low adj pressure plasma-enhanced plasma adj enhanced semi-atmosphere semi adj atmosphere high adj density adj plasma rapid adj thermal ultra-high adj vacuum ultra adj high adj vacuum limited adj reaction adj processing metalorganic) adj (CVD)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:31
S52	1481	S48 S51	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 17:32
S53	62	S52 and ((Gate adj Dielectric Gate) adj3 ("HfO2" "HfO.sub.2" "ZrO2" "ZrO.sub.2" "La2O3" "La.sub.2O.sub.3" "AlzO3" "Al.sub.zO.sub.3" "TiO2" "TiO.sub.2" "SrTiO3" "SrTiO.sub.3" "LaAlO3"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2016/09/13 17:32

		"LaAlO.sub.3" "Y2O3" "Y.sub.2O.sub.3" "HfOxNy" "HfO.sub.xN.sub.y" "ZrOxNy" "ZrO.sub.xN.sub.y" "La2OxNy" "La.sub.2O.sub.xN.sub.y" "AlzOxNy" "Al.sub.zO.sub.xN.sub.y" "TiOxNy" "TiO.sub.xN.sub.y" "SrTiOxNy" "SrTiO.sub.xN.sub.y" "LaAlOxNy" "LaAlO.sub.xN.sub.y" "Y2OxNy" "Y.sub.2O.sub.xN.sub.y" "SiON" "SiNx" "SiN.sub.x" Silicate))	DERWENT; IBM_TDB			
S54	5	S52 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:32
S55	29	S52 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:32
S57	1481	S52 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:39
S58	1452	S52 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:39
S59	318	S52 and ((Drain and Gate and Source and Spacer) with (Trench Open Cavity Contact adj Open Contact Metal ((Metal Contact) adj Liner)) with (Dielectric Isolat\$3 Liner Mask Oxide))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:40
S60	318	S57 S58 S59	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 17:40
S61	16	("20010035558" "20050124128" "20080171411" "20080272357" "20080303068" "20100075474" "6087209" "6093243" "6440828" "6812082" "7259110" "7736964" "7755114" "7785947").PN. OR ("8304819").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:07
S62	21	("20040056281" "20040121590" "20050020093" "20050026380"	US-PGPUB; USPAT;	OR	ON	2016/09/14 14:15

		"20060141712" "20060166494" "20070042580" "20070141798" "20090134497" "20110084320" "20120202326" "20120248550" "20140015104" "6323519" "6730570" "7087515" "7220668" "7348283" "7679192" "8304819" "8492217").PN. OR ("9252019").URPN.	USOCR			
S63	2	"20130049219"	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:43
S64	21	("20040056281" "20040121590" "20050020093" "20050026380" "20060141712" "20060166494" "20070042580" "20070141798" "20090134497" "20110084320" "20120202326" "20120248550" "20140015104" "6323519" "6730570" "7087515" "7220668" "7348283" "7679192" "8304819" "8492217").PN. OR ("9252019").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:44
S65	13	("20040067658" "20040150075" "20050110153" "20060003577" "6878620" "7087515" "7199044" "7253097" "7253524" "7309658" "7466027").PN. OR ("7679192").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:49
S66	4	"20130049219" "20070152334"	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:51
S67	1	"14729298"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S68	49	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S69	368	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S70	105	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S71	134	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS;	OR	ON	2016/09/14 15:09

			EPO; JPO; DERWENT; IBM_TDB			
S72	105	((Emre) near2 (Alptekin)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S73	245794	("GLOBALFOUNDRIES INC." GLOBALFOUNDRIES "IBM" International adj Business adj Machines "International Business Machines").AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S74	13131	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S75	276	("8278200").PN. ("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN. "7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502" (US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US- 20130009217-\$ or US-20090289302-\$ or US-20140145319-\$ or US- 20140138837-\$ or US-20090302480-\$ or US-20100078771-\$ or US- 20050085064-\$ or US-20060246656-\$ or US-20120132966-\$) .did. ("20070059878" "7253071" "8101518" "20070181954" "20070093047" "20020111021" "6346465" "20130119483" "20070134898" "6468901" "20080121950" "20100224938" "20120074501" "20070202692" "8338239").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S76	259027	S67 S68 S69 S70 S71 S72 S73 S74 S75	US-PGPUB; USPAT;	OR	ON	2016/09/14 15:09

			USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			
S77	10581	S76 and ((Drain Gate Source Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/14 15:09
S78	29272	S76 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Insulat\$3 Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S79	6783	S76 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S80	2155	S77 S78 S79	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/14 15:09
S81	1709	S80 and ((silicon "Si" germanium "Ge" silicon adj germanium "SiGe" "Si.sub" adj "Ge.sub" silicon adj carbide "SiC" "III-V") near3 (Substrate submount wafer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S82	1853	S80 and ((chemical adj vapor adj deposition "chemical vapor deposition" (CVD) "low-pressure CVD" (LPCVD) "plasma-enhanced CVD" (PECVD) "semi-atmosphere CVD" (SACVD) "high density plasma CVD" "high density plasma" (HDPCVD) "rapid thermal CVD" (RTCVD) "ultra-high vacuum CVD" (UHVCVD) "limited reaction processing CVD" (LRPCVD) "metalorganic CVD" (MOCVD) ((sputtering ion adj beam electron adj beam laser adj assisted physical adj vapor atomic adj layer) adj (Deposit deposition)) "electron beam deposition" "sputtering deposition" "ion beam deposition" "laser assisted deposition" "thermal oxidation" "thermal nitridation spin-on" "physical vapor deposition" (PVD) "atomic layer deposition" (ALD) "chemical oxidation" "molecular beam epitaxy" (MBE) plat\$3 evaporat\$3) ((low-pressure low adj	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09

		pressure plasma-enhanced plasma adj enhanced semi-atmosphere semi adj atmosphere high adj density adj plasma rapid adj thermal ultra-high adj vacuum ultra adj high adj vacuum limited adj reaction adj processing metalorganic) adj (CVD)))				
S83	1481	S81 S82	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/14 15:09
S84	34	S83 and (("W" Tungsten "Ti" Titanium "Ta" Tantalum "Mo" Molybdenum "TiN" Titanium adj Nitride "TaN" Tantalum adj Nitride Titanium-Tungsten Titanium adj Tungsten) with ("W" Tungsten "Co" Colbalt "Cu" Copper and "Al" Aluminum) with ("NiPt" Nickel-Platinum Nickel adj Platinum "Ti/TiN" bilayer Liner/Barrier Liner adj2 Barrier))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S85	8119	S77 S78	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/14 17:08
S86	2155	S77 S78 S79	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/14 17:08
S87	407	S76 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) with (Remov\$3 Extract\$3 Etch\$3) with (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 17:35
S88	45	S86 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) with (Remov\$3 Extract\$3 Etch\$3) with (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 17:37
S89	0	S86 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) adj5 (Remov\$3 Extract\$3 Etch\$3) adj5 (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 20:05
S90	118	S86 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse	US-PGPUB; USPAT; USOCR;	OR	ON	2016/09/14 20:06

		Diffus\$3 React\$3))) adj5 (Source Drain))	FPRS; EPO; JPO; DERWENT; IBM_TDB			
S91	33	S90 and ((Remov\$3 Extract\$3 Etch\$3) adj5 (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 20:06
S92	1480	S80 and (chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP) "laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 21:52
S93	22	(US-20150221740-\$ or US- 20130240990-\$ or US-20100078771-\$ or US-20090023261-\$ or US- 20120161324-\$ or US-20130119483-\$ or US-20140319620-\$ or US- 20140246698-\$ or US-20120104498-\$ or US-20080040697-\$ or US- 20060163631-\$ or US-20070122956-\$ or US-20130049219-\$ or US- 20070152334-\$).did. or (US-8304819-\$ or US-9224675-\$ or US-8742510-\$ or US-9093425-\$ or US-9252019-\$ or US- 6440828-\$ or US-7679192-\$).did. or (US-20110084320-\$).did.	US-PGPUB; USPAT; DERWENT	OR	ON	2016/09/14 21:52
S94	16	S93 and (chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP) "laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 21:52
S95	4	S93 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) and ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 21:57
S96	22455	H01L21/7684 H01L21/76841 H01L21/76843 H01L21/76846 H01L21/7685 H01L21/76879 H01L21/76883 H01L23/481 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 10:32
S97	69	"20150187896" "5565708"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 10:33
S98	639	S96 and ((chemical adj mechanical adj	US-PGPUB;	OR	ON	2016/09/15

		polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) and ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			12:44
S99	21	S98 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:44
S100	1	"14729298"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S101	49	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S102	368	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S103	105	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S104	134	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S105	105	((Emre) near2 (Alptekin)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S106	245815	("GLOBALFOUNDRIES INC." GLOBALFOUNDRIES "IBM" International adj Business adj Machines "International Business Machines").AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT;	OR	ON	2016/09/15 12:45

			IBM_TDB			
S107	13151	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S108	276	("8278200").PN. ("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN. "7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502" (US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US- 20130009217-\$ or US-20090289302-\$ or US-20140145319-\$ or US- 20140138837-\$ or US-20090302480-\$ or US-20100078771-\$ or US- 20050085064-\$ or US-20060246656-\$ or US-20120132966-\$) .did. ("20070059878" "7253071" "8101518" "20070181954" "20070093047" "20020111021" "6346465" "20130119483" "20070134898" "6468901" "20080121950" "20100224938" "20120074501" "20070202692" "8338239").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S109	259067	S100 S101 S102 S103 S104 S105 S106 S107 S108	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S110	19	S109 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) adj5 (Remov\$3 Extract\$3 Etch\$3) adj5 (Source Drain)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:57
S111	20386	S109 and (chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP) "laser assisted	US-PGPUB; USPAT; USOCR; FPRS;	OR	ON	2016/09/15 13:52

		deposition" Laser adj Anneal\$3 "Laser Anneal")	EPO; JPO; DERWENT; IBM_TDB			
S112	4016	S109 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) and ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 13:53
S113	0	S98 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) adj5 (Remov\$3 Extract\$3 Etch\$3) adj5 (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 13:54
S114	0	S112 and ((First Second Top Bottom Upper Lower High\$3 Low\$3 Left Right Adjacent) adj (refractory adj (material metal)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:00
S115	84	S109 and ((First Second Top Bottom Upper Lower High\$3 Low\$3 Left Right Adjacent) adj (refractory adj (material metal)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:01
S116	0	S98 and ((First Second Top Bottom Upper Lower High\$3 Low\$3 Left Right Adjacent) adj (refractory adj (material metal)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:01
S117	34	("20020019107" "20020102810" "20040206620" "20040259308" "20050087879" "20050263890" "20070145525" "20080073708" "20080138948" "20080188055" "20080272421" "20100001325" "20100025853" "20100187655" "20100213571" "20110057292" "20130011990" "5981331" "6219125" "6511879" "6593616" "6746916" "6753618" "6881642" "6906420" "7071054" "7268072" "7304367" "7329939" "7368343" "7545009" "7772678" "8105891").PN. OR ("8835274").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/15 14:04
S118	2	"20140027883"	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/15 14:04
S119	0	S115 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) and ("laser assisted deposition" Laser adj Anneal\$3 "Laser	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2016/09/15 14:11

		Anneal" Laser))	DERWENT; IBM_TDB			
S120	0	S115 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) with ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:12
S121	36	S115 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:12
S122	4	"14729298" "15826939"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/08/21 23:19
S123	1	"15826939"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/08/21 23:21
S124	2	("14729298").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/12/10 01:16
S125	76	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S126	398	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S127	148	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S128	144	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR;	OR	ON	2018/12/10 01:16

			FPRS; EPO; JPO; DERWENT; IBM_TDB			
S129	152	((Emre) near2 (Alptekin)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S130	14008	("GLOBALFOUNDRIES INC." GLOBALFOUNDRIES).AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S131	55	("8278200").PN. ("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S132	223	"7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S133	26	(US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US- 20130009217-\$ or US-20090289302-\$ or US-20140145319-\$ or US- 20140138837-\$ or US-20090302480-\$ or US-20100078771-\$ or US- 20050085064-\$ or US-20060246656-\$ or US-20120132966-\$) .did.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S134	19325	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S135	33881	S124 S125 S126 S127 S128 S129 S130 S134 S132 S133	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16

S136	2654	S135 and ((opening trench via hole) with dielectric) and ((liner or barrier) with metal) and (metal with (contact or fill))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S137	5279	S135 and ((Drain Gate Source Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2018/12/10 01:16
S138	239	S135 and ((opening or trench or via or hole) with dielectric) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S139	1030	S136 S137	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2018/12/10 01:16
S140	69	S139 and ((opening or trench or via or hole) with dielectric) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S141	14611	((Drain and Gate and Source and Spacer) with (Trench Open Cavity Contact adj Open Contact Metal ((Metal Contact) adj Liner)) with (Dielectric Isolat\$3 Liner Mask Oxide))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S142	1119	S135 and ((Drain and Gate and Source and Spacer) with (Trench Open Cavity Contact adj Open Contact Metal ((Metal Contact) adj Liner)) with (Dielectric Isolat\$3 Liner Mask Oxide))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S143	150	S135 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S144	31	S142 S143	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	AND	ON	2018/12/10 01:16

			DERWENT; IBM_TDB			
S145	201124	((Gate Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2018/12/10 01:16
S146	77111	S145 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S147	25930	S145 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S148	1264	S145 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S149	2700	S145 and (("W" Tungsten "Ti" Titanium "Ta" Tantalum "Mo" Molybdenum "TiN" Titanium adj Nitride "TaN" Tantalum adj Nitride Titanium-Tungsten Titanium adj Tungsten) with ("W" Tungsten "Co" Colbalt "Cu" Copper and "Al" Aluminum) with ("NiPt" Nickel-Platinum Nickel adj Platinum "Ti/TiN" bilayer Liner/Barrier Liner adj2 Barrier))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S150	4	"14729298"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S151	76	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S152	398	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S153	148	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB;	OR	ON	2018/12/10

			USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			01:16
S154	144	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S155	152	((Emre) near2 (Alptekin)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S156	267263	("GLOBALFOUNDRIES INC." GLOBALFOUNDRIES "IBM" International adj Business adj Machines "International Business Machines").AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S157	19325	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S158	320	("8278200").PN. ("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN. "7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502" (US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US- 20130009217-\$ or US-20090289302-\$ or US-20140145319-\$ or US- 20140138837-\$ or US-20090302480-\$ or US-20100078771-\$ or US- 20050085064-\$ or US-20060246656-\$ or US-20120132966-\$) .did.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16

		("20070059878" "7253071" "8101518" "20070181954" "20070093047" "20020111021" "6346465" "20130119483" "20070134898" "6468901" "20080121950" "20100224938" "20120074501" "20070202692" "8338239").PN.				
S159	286304	S150 S151 S152 S153 S154 S155 S156 S157 S158	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S160	13909	S159 and ((Drain Gate Source Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2018/12/10 01:16
S161	36667	S159 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Insulat\$3 Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S162	455	S159 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S163	397	S159 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S164	8346	S159 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S165	10782	S160 S161	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2018/12/10 01:16
S166	2752	S160 S161 S164	US-PGPUB; USPAT; USOCR; FPRS;	AND	ON	2018/12/10 01:16

			EPO; JPO; DERWENT; IBM_TDB			
S167	14	S162 S163	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2018/12/10 01:16
S168	6	S166 S167	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2018/12/10 01:16
S169	55	S166 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S170	39	S166 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S171	2172	S166 and ((silicon "Si" germanium "Ge" silicon adj germanium "SiGe" "Si.sub" adj "Ge.sub" silicon adj carbide "SiC" "III-V") near3 (Substrate submount wafer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S172	101	S166 and ((Gate adj Dielectric Gate) adj3 ("HfO2" "HfO.sub.2" "ZrO2" "ZrO.sub.2" "La2O3" "La.sub.2O.sub.3" "AlzO3" "Al.sub.zO.sub.3" "TiO2" "TiO.sub.2" "SrTiO3" "SrTiO.sub.3" "LaAlO3" "LaAlO.sub.3" "Y2O3" "Y.sub.2O.sub.3" "HfOxNy" "HfO.sub.xN.sub.y" "ZrOxNy" "ZrO.sub.xN.sub.y" "La2OxNy" "La.sub.2O.sub.xN.sub.y" "AlzOxNy" "Al.sub.zO.sub.xN.sub.y" "TiOxNy" "TiO.sub.xN.sub.y" "SrTiOxNy" "SrTiO.sub.xN.sub.y" "LaAlOxNy" "LaAlO.sub.xN.sub.y" "Y2OxNy" "Y.sub.2O.sub.xN.sub.y" "SiON" "SiNx" "SiN.sub.x" Silicate))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S173	51383	S159 and ((chemical adj vapor adj deposition "chemical vapor deposition" (CVD) "low-pressure CVD" (LPCVD) "plasma-enhanced CVD" (PECVD) "semi-atmosphere CVD" (SACVD) "high density plasma CVD" "high density plasma" (HDPCVD) "rapid thermal CVD" (RTCVD) "ultra-high vacuum CVD"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16

		(UHVCVD) "limited reaction processing CVD" (LRPCVD) "metalorganic CVD" (MOCVD) ((sputtering ion adj beam electron adj beam laser adj assisted physical adj vapor atomic adj layer) adj (Deposit deposition)) "electron beam deposition" "sputtering deposition" "ion beam deposition" "laser assisted deposition" "thermal oxidation" "thermal nitridation spin-on" "physical vapor deposition" (PVD) "atomic layer deposition" (ALD) "chemical oxidation" "molecular beam epitaxy" (MBE) plat\$3 evaporat\$3) ((low-pressure low adj pressure plasma-enhanced plasma adj enhanced semi-atmosphere semi adj atmosphere high adj density adj plasma rapid adj thermal ultra-high adj vacuum ultra adj high adj vacuum limited adj reaction adj processing metalorganic) adj (CVD)))				
S174	2414	S166 and ((chemical adj vapor adj deposition "chemical vapor deposition" (CVD) "low-pressure CVD" (LPCVD) "plasma-enhanced CVD" (PECVD) "semi-atmosphere CVD" (SACVD) "high density plasma CVD" "high density plasma" (HDPCVD) "rapid thermal CVD" (RTCVD) "ultra-high vacuum CVD" (UHVCVD) "limited reaction processing CVD" (LRPCVD) "metalorganic CVD" (MOCVD) ((sputtering ion adj beam electron adj beam laser adj assisted physical adj vapor atomic adj layer) adj (Deposit deposition)) "electron beam deposition" "sputtering deposition" "ion beam deposition" "laser assisted deposition" "thermal oxidation" "thermal nitridation spin-on" "physical vapor deposition" (PVD) "atomic layer deposition" (ALD) "chemical oxidation" "molecular beam epitaxy" (MBE) plat\$3 evaporat\$3) ((low-pressure low adj pressure plasma-enhanced plasma adj enhanced semi-atmosphere semi adj atmosphere high adj density adj plasma rapid adj thermal ultra-high adj vacuum ultra adj high adj vacuum limited adj reaction adj processing metalorganic) adj (CVD)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S175	1916	S171 S174	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2018/12/10 01:16
S176	76	S175 and ((Gate adj Dielectric Gate) adj3 ("HfO2" "HfO.sub.2" "ZrO2" "ZrO.sub.2" "La2O3" "La.sub.2O.sub.3" "Al2O3" "Al.sub.2O.sub.3" "TiO2" "TiO.sub.2" "SrTiO3" "SrTiO.sub.3" "LaAlO3" "LaAlO.sub.3" "Y2O3"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT;	OR	ON	2018/12/10 01:16

		"Y.sub.2O.sub.3" "HfOxNy" "HfO.sub.xN.sub.y" "ZrOxNy" "ZrO.sub.xN.sub.y" "La2OxNy" "La.sub.2O.sub.xN.sub.y" "AlzOxNy" "Al.sub.zO.sub.xN.sub.y" "TiOxNy" "TiO.sub.xN.sub.y" "SrTiOxNy" "SrTiO.sub.xN.sub.y" "LaAlOxNy" "LaAlO.sub.xN.sub.y" "Y2OxNy" "Y.sub.2O.sub.xN.sub.y" "SiON" "SiNx" "SiN.sub.x" Silicate))	IBM_TDB			
S177	23	S175 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S178	32	S175 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S179	1916	S175 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S180	1883	S175 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S181	471	S175 and ((Drain and Gate and Source and Spacer) with (Trench Open Cavity Contact adj Open Contact Metal ((Metal Contact) adj Liner)) with (Dielectric Isolat\$3 Liner Mask Oxide))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S182	471	S179 S180 S181	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2018/12/10 01:16
S183	21	("20010035558" "20050124128" "20080171411" "20080272357" "20080303068" "20100075474" "6087209" "6093243" "6440828" "6812082" "7259110" "7736964" "7755114" "7785947").PN. OR ("8304819").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2018/12/10 01:16
S184	21	("20040056281" "20040121590" "20050020093" "20050026380"	US-PGPUB; USPAT;	OR	ON	2018/12/10 01:16

		"20060141712" "20060166494" "20070042580" "20070141798" "20090134497" "20110084320" "20120202326" "20120248550" "20140015104" "6323519" "6730570" "7087515" "7220668" "7348283" "7679192" "8304819" "8492217").PN. OR ("9252019").URPN.	USOCR			
S185	2	"20130049219"	US-PGPUB; USPAT; USOCR	OR	ON	2018/12/10 01:16
S186	21	("20040056281" "20040121590" "20050020093" "20050026380" "20060141712" "20060166494" "20070042580" "20070141798" "20090134497" "20110084320" "20120202326" "20120248550" "20140015104" "6323519" "6730570" "7087515" "7220668" "7348283" "7679192" "8304819" "8492217").PN. OR ("9252019").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2018/12/10 01:16
S187	15	("20040067658" "20040150075" "20050110153" "20060003577" "6878620" "7087515" "7199044" "7253097" "7253524" "7309658" "7466027").PN. OR ("7679192").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2018/12/10 01:16
S188	4	"20130049219" "20070152334"	US-PGPUB; USPAT; USOCR	OR	ON	2018/12/10 01:16
S189	4	"14729298"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S190	76	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S191	398	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S192	148	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S193	144	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS;	OR	ON	2018/12/10 01:16

			EPO; JPO; DERWENT; IBM_TDB			
S194	152	((Emre) near2 (Alptekin)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S195	267263	("GLOBALFOUNDRIES INC." GLOBALFOUNDRIES "IBM" International adj Business adj Machines "International Business Machines").AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S196	19325	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S197	320	("8278200").PN. ("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN. "7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502" (US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US- 20130009217-\$ or US-20090289302-\$ or US-20140145319-\$ or US- 20140138837-\$ or US-20090302480-\$ or US-20100078771-\$ or US- 20050085064-\$ or US-20060246656-\$ or US-20120132966-\$) .did. ("20070059878" "7253071" "8101518" "20070181954" "20070093047" "20020111021" "6346465" "20130119483" "20070134898" "6468901" "20080121950" "20100224938" "20120074501" "20070202692" "8338239").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S198	286304	S189 S190 S191 S192 S193 S194 S195 S196 S197	US-PGPUB; USPAT;	OR	ON	2018/12/10 01:16

			USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			
S199	13909	S198 and ((Drain Gate Source Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2018/12/10 01:16
S200	36667	S198 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Insulat\$3 Isolot\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S201	8346	S198 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S202	2752	S199 S200 S201	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2018/12/10 01:16
S203	2172	S202 and ((silicon "Si" germanium "Ge" silicon adj germanium "SiGe" "Si.sub" adj "Ge.sub" silicon adj carbide "SiC" "III-V") near3 (Substrate submount wafer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16
S204	2414	S202 and ((chemical adj vapor adj deposition "chemical vapor deposition" (CVD) "low-pressure CVD" (LPCVD) "plasma-enhanced CVD" (PECVD) "semi-atmosphere CVD" (SACVD) "high density plasma CVD" "high density plasma" (HDPCVD) "rapid thermal CVD" (RTCVD) "ultra-high vacuum CVD" (UHVCVD) "limited reaction processing CVD" (LRPCVD) "metalorganic CVD" (MOCVD) ((sputtering ion adj beam electron adj beam laser adj assisted physical adj vapor atomic adj layer) adj (Deposit deposition)) "electron beam deposition" "sputtering deposition" "ion beam deposition" "laser assisted deposition" "thermal oxidation" "thermal nitridation spin-on" "physical vapor deposition" (PVD) "atomic layer deposition" (ALD) "chemical oxidation" "molecular beam epitaxy" (MBE) plat\$3 evaporat\$3) ((low-pressure low adj	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/10 01:16

		pressure plasma-enhanced plasma adj enhanced semi-atmosphere semi adj atmosphere high adj density adj plasma rapid adj thermal ultra-high adj vacuum ultra adj high adj vacuum limited adj reaction adj processing metalorganic) adj (CVD))				
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12/ 25/ 2018 10:01:28 PM
C:\ Users\ Inguyen16\ Documents\ EAST\ Workspaces\ 15826939.wsp

Bibliographic Data

Application No: 15/826,939

Foreign Priority claimed: ☐ Yes ☒ No

35 USC 119 (a-d) conditions met: ☐ Yes ☒ No

☐ Met After Allowance

Verified and Acknowledged:

/LAWRENCE-LIN T
NGUYEN/

Examiner's Signature

Initials

Title:

CONTACTS TO SEMICONDUCTOR SUBSTRATE AND
METHODS OF FORMING SAME

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
11/30/2017	257	2897	FIS920150066US2
RULE			

APPLICANTS

GLOBALFOUNDRIES INC., Grand Cayman, CAYMAN ISLANDS

INVENTORS

Emre Alptekin Wappingers Falls, NY, UNITED STATES

Nicolas L. Breil San Jose, CA, UNITED STATES

Christian Lavoie Pleasantville, NY, UNITED STATES

Ahmet S. Ozcan Chappaqua, NY, UNITED STATES

Kathryn T. Schonenberg Wappingers Falls, NY, UNITED STATES

CONTINUING DATA

This application is a DIV of 14729298 06/03/2015 PAT 9865546

FOREIGN APPLICATIONS

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For the following listed application(s), please recognize as the "Fee Address" under the provisions of 37 CFR 1.363 the address associated with:



Customer Number:

00197**OR**

The attached Request for Customer Number (PTO/SB/125) form.

PATENT NUMBER (if known)	APPLICATION NUMBER
	15/826,939

Completed by (check one):



Applicant/Inventor

/John A. Merecki/

Signature



Attorney or Agent of record 35,812
 (Reg. No.)

John A. Merecki

Typed or printed name



Assignee of record of the entire interest. See 37 CFR 3.71.
 Statement under 37 CFR 3.73(b) is enclosed.
 (Form PTO/SB/96)

518-449-0044

Requester's telephone number



Assignee recorded at Reel _____ Frame _____

May 29, 2020

Date

NOTE: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required. Submit multiple forms if more than one signature is required, see below*.

* Total of 1 forms are submitted.

This collection of information is required by 37 CFR 1.363. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 5 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND COMPLETE D FORMS TO THIS ADDRESS.**
SEND TO: Mail Stop M Correspondence, Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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Electronic Patent Application Fee Transmittal

Application Number:	15826939			
Filing Date:	30-Nov-2017			
Title of Invention:	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME			
First Named Inventor/Applicant Name:	Emre Alptekin			
Filer:	John Alexander Merecki/Michele Mathews			
Attorney Docket Number:	FIS920150066US2			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
UTILITY APPL ISSUE FEE	1501	1	1000	1000

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1000

Electronic Acknowledgement Receipt

EFS ID:	39571348
Application Number:	15826939
International Application Number:	
Confirmation Number:	6013
Title of Invention:	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME
First Named Inventor/Applicant Name:	Emre Alptekin
Customer Number:	133964
Filer:	John Alexander Merecki/Michele Mathews
Filer Authorized By:	John Alexander Merecki
Attorney Docket Number:	FIS920150066US2
Receipt Date:	29-MAY-2020
Filing Date:	30-NOV-2017
Time Stamp:	13:20:16
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 1000
RAM confirmation Number	E20205SD21109028
Deposit Account	500999
Authorized User	Michele Mathews

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37 CFR 1.20 (Post Issuance fees)

37 CFR 1.21 (Miscellaneous fees and charges)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	FIS920150066US2_IssueFeeTransmittal_29MAY2020.pdf	123294	no	1
			a444a2e21f2253dc04aeff01b9c868049e1fe859		
Warnings:					
Information:					
2	Maintenance Fee Address Change	FIS920150066US2_FeeAddress Form_29MAY2020.pdf	312543	no	2
			36d966cec521a7f6cb4bbe951d7ae5853b97c7e3		
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	30693	no	2
			7e3c076748390cf97e6067dfc7b1abd24c865d06		
Warnings:					
Information:					
Total Files Size (in bytes):			466530		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

133964 7590 03/23/2020
HOFFMAN WARNICK LLC
540 BROADWAY
4TH FLOOR
ALBANY, NY 12207

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/826.939	11/30/2017	Emre Alptekin	FIS920150066US2	6013

TITLE OF INVENTION: CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0.00	\$0.00	\$1000	06/23/2020

EXAMINER	ART UNIT	CLASS-SUBCLASS
TOLEDO, FERNANDO L	2897	257-757000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☒ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,
(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 Anthony Canale
2 Hoffman Warnick LLC
3

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

GLOBALFOUNDRIES INC.

Grand Cayman, KY

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☒ Issue Fee ☒ Publication Fee (if required) ☐ Advance Order - # of Copies

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☒ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. 500999

5. Change in Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
☐ Applicant asserting small entity status. See 37 CFR 1.27
☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.
NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.
NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature /John A. Merecki/

Date May 29, 2020

Typed or printed name John A. Merecki

Registration No. 35,812



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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/826,939	07/07/2020	10707167	FIS920150066US2	6013

133964 7590 06/17/2020
HOFFMAN WARNICK LLC
540 BROADWAY
4TH FLOOR
ALBANY, NY 12207

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 362 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

GLOBALFOUNDRIES INC., Grand Cayman, CAYMAN ISLANDS;
Emre Alptekin, Wappingers Falls, NY;
Nicolas L. Breil, San Jose, CA;
Christian Lavoie, Pleasantville, NY;
Ahmet S. Ozcan, Chappaqua, NY;
Kathryn T. Schonenberg, Wappingers Falls, NY;

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